



Review of generative AI for synthetic data generation: a healthcare perspective

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Received: 23 February 2025 / Accepted: 30 October 2025 / Published online: 10 December 2025
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Abstract

Generative AI has emerged as a transformative technology in healthcare, enabling the generation of high-fidelity synthetic data for applications such as medical imaging, electronic health records, biomedical signal processing, and drug discovery. The increasing reliance on machine learning in healthcare necessitates large-scale, high-quality datasets, yet real-world data acquisition is often constrained by privacy regulations, heterogeneity, and limited accessibility. Generative AI models provide a viable solution by generating realistic and diverse synthetic datasets while preserving patient confidentiality. Unlike prior reviews that primarily focus on specific model classes or applications, this study fills a significant research gap by offering a unified, comparative evaluation of diverse generative models, including Generative Adversarial Networks, Variational Autoencoders, Transformers, and Diffusion Models, as well as their adaptations for privacy-preserving Federated Learning environments. Each model class is examined in terms of its variants, underlying methodologies, performance in healthcare applications, strengths, limitations, and computational feasibility. The study also investigates practical considerations for deploying generative AI in clinical settings, including challenges related to training stability, bias mitigation, model interpretability, and regulatory compliance. The insights from this review provide guidance for researchers and healthcare practitioners in selecting and optimizing generative AI models for medical applications, laying the foundation for future advancements in AI-driven healthcare solutions.

Keywords Generative AI · Synthetic data generation · Healthcare datasets · Privacy-preserving AI · Federated learning · Differential privacy

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1 Introduction

Generative AI has emerged as a transformative technology in healthcare, particularly in synthetic data generation, addressing key challenges such as data privacy, accessibility, and bias (van Breugel et al. 2024; Raza et al. 2024; Kaabachi et al. 2025). The healthcare industry relies heavily on large-scale, high-quality datasets for developing AI-driven diagnostic tools, treatment recommendations, and predictive analytics (Yan et al. 2022). However, obtaining real-world medical data is often constrained by strict privacy regulations, ethical concerns, patient consent limitations, and institutional data limitations (Jonnagaddala and Wong 2025). Additionally, real-world medical datasets often suffer from imbalanced distributions, underrepresenting specific demographics, and data inconsistencies across institutions, leading to biased AI models (Salmi et al. 2024; Arora et al. 2023). These challenges necessitate advanced synthetic data generation techniques capable of producing realistic, diverse, and privacy-preserving medical datasets. Generative AI plays a vital role in this domain, enabling the generation of high-fidelity synthetic data that can be leveraged for model training, data augmentation, and enhanced generalization in healthcare applications without compromising sensitive patient information (Giuffrè and Shung 2023; Sella et al. 2025; Zhang et al. 2025; Zhou et al. 2024b).

The development of generative models has introduced diverse techniques for synthetic data generation, each designed to handle different healthcare data modalities. Generative Adversarial Networks (GANs) (Goodfellow et al. 2014) are among the most widely adopted techniques, known for their ability to produce high-resolution synthetic medical images by learning complex data distributions (Chakraborty et al. 2024; Farajzadeh-Zanjani et al. 2022; Jabbar et al. 2021). They have played a key role in enhancing radiological image synthesis, histopathological data augmentation, and domain adaptation across imaging modalities such as MRI, CT, and ultrasound. Variational Autoencoder-based models, including standard Variational Autoencoders (VAEs) (Kingma and Welling 2014) and Convolutional Variational Autoencoders (CVAEs), provide robust solutions for structured data synthesis, denoising, anomaly detection, and feature extraction, making them valuable in applications such as Electronic Health Records (EHR) augmentation and genomic data modeling (Chen and Guo 2023; Berahmand et al. 2024; Pawar and Attar 2020; Wei and Mahmood 2020). Transformers, particularly Generative Pre-trained Transformers (GPTs) and Vision Transformers (ViTs), have demonstrated exceptional capabilities in medical text generation, diagnostic report automation, and multimodal AI integration, bridging the gap between clinical language processing and imaging-based diagnosis (Yenduri et al. 2024; Zhang and Shao 2024; Islam et al. 2023; Parvaiz et al. 2023; Goyal et al. 2023; Han et al. 2022).

Another significant advancement in generative AI for healthcare is the emergence of Diffusion Models (Ho et al. 2020), which have demonstrated state-of-the-art performance in high-fidelity synthetic data generation. Unlike GANs, which often struggle with training instability and mode collapse, Diffusion models leverage probabilistic modeling techniques to generate structured, controllable, and interpretable synthetic data (Tashiro et al. 2021; Wang et al. 2023c). These models have shown remarkable effectiveness in applications such as MRI reconstruction, histopathological image enhancement, and structured EHR synthesis. Variants such as Latent Diffusion Models (LDMs) and Conditional Diffusion Models (CDMs) provide targeted control over data synthesis, ensuring higher clinical validity and consistency in generated medical datasets (Pinaya et al. 2022; Zhu et al. 2023). Although

Diffusion Models require extensive computational resources and longer inference times, recent optimizations such as Progressive Distillation Diffusion Models (PDDMs) aim to accelerate the generation process (Zhou et al. 2024a), making them more viable for real-time clinical applications.

With increasing concerns about data privacy and regulatory compliance, Federated Learning (FL) has emerged as a paradigm shift in generative AI, allowing multiple healthcare institutions to collaboratively train models without sharing raw patient data (Little et al. 2023; Chen and Pan 2024). Federated Generative Models, such as Federated GANs and Federated VAEs, facilitate decentralized learning, enabling privacy-preserving AI applications while ensuring secure multi-institutional collaborations (Yang et al. 2023). These models are particularly valuable in fields such as medical imaging, real-time patient monitoring, and multi-source biomedical data synthesis, mitigating concerns related to data centralization and accessibility restrictions. While FL enables institutions to collaboratively train models without exchanging raw patient data, it still faces challenges such as data heterogeneity, limited local sample sizes, and class imbalance across different sites. Generative modeling complements FL by synthesizing additional high-fidelity data locally at each node, thus enhancing model generalization, mitigating imbalance, and reducing training bias. Moreover, synthetic data can be used to augment rare cases or simulate underrepresented patient subpopulations, which are critical for ensuring robust and equitable AI models across federated systems (Guo et al. 2024).

The motivation behind this review stems from the growing fragmentation of literature across different generative AI models in healthcare. To ensure a comprehensive and up-to-date review, we selected articles published between January 2018 and April 2025, with most included papers from 2023 and 2024. Literature was sourced using targeted healthcare-related generative AI keywords across Google Scholar, PubMed, IEEE Xplore, ACM Digital Library, and arXiv. Existing surveys often specialize in single techniques (e.g., GANs or VAEs) or restrict themselves to specific applications such as medical imaging. However, there remains a critical need for a comprehensive, comparative, and methodologically diverse review that not only discusses the state-of-the-art generative models but also examines their integration with privacy-preserving frameworks such as FL. This manuscript addresses this gap by presenting a comprehensive review and analysis of four major categories of generative AI models for synthetic data generation, such as GANs, VAEs, Transformers, and Diffusion Models, alongside their respective variants. Additionally, we also explore several generative models in FL environments for synthetic data generation. This review comprehensively discusses methodologies of the generative models, their strengths, weaknesses, and applicability across healthcare domains, including medical imaging, EHR processing, genomic data modeling, and physiological signal synthesis. Key aspects such as model architecture, training complexity, data fidelity, generalization capabilities, and computational efficiency are critically examined to provide a holistic understanding of their practical implementations. Additionally, this study delves into the limitations of each model, highlighting challenges related to interpretability, ethical concerns, domain adaptation, and real-world deployment feasibility. By synthesizing insights from recent advancements in generative AI, this review aims to serve as a guide for researchers, clinicians, and AI practitioners in selecting the most suitable generative models for healthcare applications while ensuring optimal performance, scalability, and compliance with data protection regulations.

Beyond the comprehensive analysis, this study also explores the broader implications of generative AI in shaping the future of medical AI research, personalized healthcare, and clinical decision support systems. The role of synthetic data in enhancing AI-driven diagnostics, augmenting rare disease datasets, and reducing model bias is carefully examined. Additionally, future directions in generative AI, including hybrid model architectures, self-supervised learning approaches, and explainable AI frameworks, are discussed to address existing challenges and unlock new possibilities in healthcare AI. The findings of this study contribute to the growing discourse on the role of generative AI in healthcare, paving the way for future research and clinical integration of AI-driven synthetic data solutions that can enhance the reliability, accessibility, and efficiency of healthcare systems worldwide.

This paper is structured to provide a comprehensive understanding of generative AI models in healthcare, detailing their methodologies, applications, and challenges. Section 2 outlines the taxonomy of generative AI models, providing essential knowledge for understanding their role in healthcare synthetic data generation. Section 3 presents an in-depth review and comprehensive analysis of four major categories of generative AI models with their variants, alongside FL-based several variants, emphasizing their strengths, limitations, and applications in healthcare domains. Section 4 discusses the performance of these models, highlighting their suitability for different healthcare data types, training complexities, and real-world deployment challenges. Finally, the paper summarizes key findings, discusses ongoing limitations, and proposes future research directions for advancing generative AI in healthcare applications.

2 Taxonomy of generative AI models

This section first describes the methods used to identify, screen, and extract studies included in this survey. We provide details of the scope, keyword searches, eligibility criteria, screening procedures, quality assessment, and data extraction, complemented by Table 1, which summarizes the database-specific queries and filters. As this work is a structured survey rather than a formal systematic review, we did not compute exclusion counts at each step. Instead, we present a transparent account of our search strategy and assessment rubric to enhance reproducibility. After outlining these methodological steps, we introduce a taxonomy of generative models relevant to healthcare. This taxonomy distinguishes training paradigms (such as adversarial, variational, and diffusion-based learning) from flexible architectures (e.g., Transformers) and also considers privacy-preserving frameworks such as FL.

2.1 Method

Scope and Sources: We reviewed literature on generative models for synthetic healthcare data published between 1 January 2018 and 20 April 2025. The databases searched included PubMed, IEEE Xplore, ACM Digital Library, arXiv, and Google Scholar, covering biomedical, engineering, and computer science fields. This time window was chosen to capture modern architectures (GANs, VAEs, Diffusion Models, Transformers) and privacy-preserving extensions (such as differential privacy, FL).

Table 1 Search strategy—keywords and database-specific filters used for literature retrieval (January 2018–April 2025)

Domain	Databases	Keyword search (representative terms)	Filters applied
General – Healthcare Synthetic Data	PubMed, IEEE Xplore, ACM DL, arXiv, Google Scholar	synthetic data, data synthesis, generative, healthcare, clinical, patient, GAN, generative adversarial, VAE, variational autoencoder, diffusion, transformer, GPT, latent diffusion	Year: 2018–2025; English language; exclude patents and citations (Google Scholar); journals, conferences, early access (IEEE/ACM)
Medical Imaging	PubMed, IEEE Xplore, ACM DL, arXiv, Google Scholar	synthetic data, data synthesis, generative, radiology, imaging, MRI, CT, X-ray, ultrasound, histopathology, pathology, dermatology, ophthalmology, retinal, endoscopy, GAN, VAE, diffusion, transformer, GPT	Same as above
EHR and Tabular Data	PubMed, IEEE Xplore, ACM DL, arXiv, Google Scholar	synthetic data, data synthesis, generative, electronic health record, EHR, tabular, longitudinal, structured data, GAN, VAE, diffusion, transformer, GPT	Same as above
Genomics and Bioinformatics	PubMed, IEEE Xplore, ACM DL, arXiv, Google Scholar	synthetic data, data synthesis, generative, genomic, genetic, sequencing, RNA-seq, transcriptomic, proteomic, bioinformatics, GAN, VAE, diffusion, transformer, GPT	Same as above
Time-Series and Physiological Signals	PubMed, IEEE Xplore, ACM DL, arXiv, Google Scholar	synthetic data, data synthesis, generative, ECG, EEG, vital sign, time series, physiological, biosignal, wearable, monitoring, GAN, VAE, diffusion, transformer, GPT	Same as above
Clinical Text and NLP	PubMed, IEEE Xplore, ACM DL, arXiv, Google Scholar	synthetic data, data synthesis, generative, clinical note, discharge summary, radiology report, medical text, NLP, natural language, biomedical language, GAN, VAE, diffusion, transformer, GPT	Same as above
Multimodal and Cross-Domain	PubMed, IEEE Xplore, ACM DL, arXiv, Google Scholar	synthetic data, data synthesis, generative, multimodal, cross-modal, cross-domain, fusion, integration, GAN, VAE, diffusion, transformer, GPT	Same as above
Federated Learning and Privacy	PubMed, IEEE Xplore, ACM DL, arXiv, Google Scholar	synthetic data, data synthesis, generative, federated, decentralized, privacy, differential privacy, secure, GAN, VAE, diffusion, transformer, GPT	Same as above

Keyword Searches: We used a combination of broad umbrella terms and domain-specific refinements tailored to healthcare modalities (imaging, EHR/tabular, genomics, time-series, clinical text, multimodal data) and privacy-preserving approaches (FL, differential privacy). The representative search strategy for each database is presented in Table 1. Queries combined terms related to generative model families (GAN, VAE, diffusion, Transformer, GPT, latent diffusion) with healthcare-relevant keywords.

Eligibility Criteria: Studies were included if they (i) proposed or evaluated generative models producing synthetic healthcare data, (ii) reported at least one utility or quality metric (e.g., FID, IS, AUROC, AUPRC, MAE, MMD) or privacy evaluation (e.g., ϵ -differential privacy, resistance to membership inference), and (iii) were published in English between

2018 and 2025. We excluded studies that (i) focused solely on discriminative models or augmentation-only approaches, (ii) were unrelated to healthcare applications, or (iii) were commentaries, editorials, or lacked full-text availability.

Screening and Selection: All records were deduplicated and screened in two stages, such as (i) title/abstract screening to exclude irrelevant, non-generative, or non-healthcare studies, and (ii) full-text eligibility assessment for remaining records. Screening was conducted independently by two reviewers, with disagreements resolved through discussion and, when necessary, arbitration by a third reviewer. Since this was a survey rather than a systematic review, exclusion numbers were not tracked at each stage.

Quality and Relevance Assessment: We did not compute numerical quality scores. Instead, we applied a structured qualitative rubric based on (i) publication venue (peer-reviewed journal/conference vs. preprint), (ii) methodological clarity (e.g., model description, dataset availability, baselines), (iii) evaluation depth (quantitative benchmarks and/or clinical validation), (iv) reproducibility indicators (e.g., code/data availability, hyperparameter reporting), and (v) citation activity at the time of screening.

Data Extraction and Synthesis: For each included study, we extracted information on the clinical domain (such as imaging, EHR, genomics, physiological signals, clinical text, multimodal), task, dataset(s), model family/variant, privacy mechanism (such as differential privacy, FL, hybrid), evaluation metrics, and training details. Given the heterogeneity of benchmarks, synthesis was primarily qualitative. Where possible, representative quantitative metrics were consolidated into comparative summaries (see Section 4.1.5, Table 13).

2.2 Taxonomy of generative models

Based on the methodological framework described above, we present a taxonomy of generative AI models relevant to healthcare. This taxonomy organizes the diverse approaches identified in our review into distinct families, based on their training paradigms and architectural characteristics.

Generative models are machine learning methods capable of synthesizing new data samples that approximate the distribution of real-world data. Unlike discriminative models, which focus on classification or prediction, generative models capture latent data structures and produce novel instances that exhibit high realism and fidelity. This ability is especially valuable in healthcare, where data scarcity, privacy constraints, and limited access to rare disease patient groups often restrict the availability of large annotated datasets. By generating synthetic yet realistic datasets, these models enable robust AI development while mitigating patient privacy concerns and aligning with regulatory frameworks.

In constructing our taxonomy (Fig. 1), we distinguish training paradigms, such as adversarial learning (GANs), variational inference (VAEs), and iterative refinement (Diffusion Models), from architectural families such as Transformers, which can be combined with multiple paradigms. We also include federated and privacy-preserving variants, which extend these families to decentralized or regulated healthcare settings.

The application of generative AI in healthcare spans multiple domains, each benefiting from synthetic data generation to improve diagnostic accuracy, model robustness, and research advancements. In medical imaging, GANs (Chen et al. 2018; Marcos et al. 2021) and Diffusion Models (Tiago et al. 2023) enhance the quality of diagnostic images, assist in image reconstruction, and enable super-resolution techniques for improved radiological

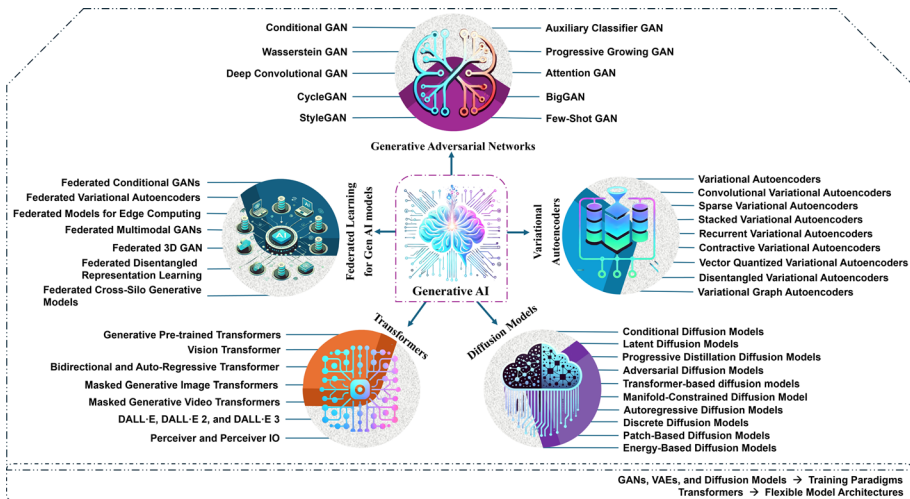


Fig. 1 Taxonomy of generative AI models for healthcare

analysis (Ngasa et al. 2024). Structured EHR synthesis using VAEs and discrete generative models ensures data augmentation while maintaining patient privacy, facilitating predictive modeling and personalized treatment planning. Generative AI also plays a crucial role in time-series forecasting. Recurrent Autoencoders and Autoregressive Diffusion Models capture temporal dependencies in patient monitoring data, improving disease progression modeling and real-time clinical decision-making.

3 Generative AI models for synthetic data generation

Generative models leverage deep learning techniques to generate high-quality medical images, structured EHRs, and time-series data, facilitating AI-driven advancements in medical research and clinical applications. Various generative architectures have been explored, including GANs, VAEs, Transformers, Diffusion Models, and FL-based generative models. Each category introduces unique mechanisms for improving data fidelity, stability, and interpretability. GANs excel in high-resolution medical image synthesis, VAEs provide structured feature learning for anomaly detection, Transformers enhance multimodal data processing, Diffusion Models offer state-of-the-art controllable data generation, and FL-based generative models enable privacy-preserving AI training.

3.1 Generative adversarial networks

Generative adversarial networks (GANs) are adversarial frameworks where a generator learns to synthesize data while a discriminator distinguishes real from synthetic samples. This adversarial dynamic enables GANs to achieve state-of-the-art realism in medical image synthesis, unpaired modality translation, and augmentation of rare disease datasets. Across healthcare, GANs are used in radiology (MRI to CT synthesis, low-dose CT denoising), histopathology (rare tumor subtype enrichment), and ophthalmology and der-

matology (fundus and lesion synthesis). Numerous variants, including conditional GANs, CycleGANs, WGANs, and StyleGANs, adapt the baseline framework to improve stability, controllability, or suitability for specific tasks. Detailed variant-level discussions, including Table 2, follow in the subsections below.

3.1.1 Deep convolutional GAN

Deep Convolutional GAN (DCGAN) is a foundational variant of GAN that leverages convolutional layers to improve the quality and stability of generative models (Liu et al. 2022b). Introduced by Radford (2015), DCGAN replaces fully connected layers in the generator and discriminator with convolutional and transposed convolutional layers, optimizing them for image-based data generation. The generator uses transposed convolutions to upscale latent vectors into realistic images, while the discriminator employs stride-based convolutions to downscale and classify inputs as real or synthetic. Both networks use batch normalization to stabilize training and prevent mode collapse and non-linear activation functions (ReLU for the generator and Leaky ReLU for the discriminator) to improve feature learning. DCGAN is particularly well-suited for tasks involving visual data due to its ability to model spatial hierarchies effectively, making it a key tool for applications in healthcare.

Table 2 Comparison of GAN Variants in Healthcare Applications

Variant	Key features	Supported data types	Limitations
Conditional GAN	Generates data conditioned on specific attributes; Improves class-specific data generation	Medical images; Structured EHR data	Training instability; Mode collapse
Wasserstein GAN	Uses Wasserstein distance for stable training; Reduces mode collapse	Medical imaging; Time-series data	Computationally expensive; Requires careful tuning
Deep Convolutional GAN	Uses convolutional layers for high-quality image synthesis; Stable architecture	Medical imaging; Histopathology slides	Struggles with complex textures and fine details
CycleGAN	Enables unpaired image-to-image translation; Preserves domain characteristics	Cross-modal medical imaging (CT-MRI, histology-stained images)	Training complexity; Artifacts in generated images
StyleGAN	Generates high-resolution images with style control; Provides disentangled representations	High-resolution radiology and pathology images	High computational cost; Requires significant fine-tuning
Auxiliary Classifier GAN	Integrates class labels into discriminator; Improves class-specific generation	Disease classification datasets; Medical scans	Prone to overfitting on small datasets
Progressive Growing GAN	Progressive training stabilizes high-resolution image generation	High-resolution medical images (CT, MRI, histopathology)	Requires large memory and computational power
Attention/Self-Attention GAN	Uses attention mechanisms for fine-grained details; Improves long-range dependencies	Medical image generation from text descriptions; Enhanced CT/MRI scans	Slower training; Requires large datasets
Big GAN	Large-scale architecture for generating highly diverse images; Class-conditional normalization	Multi-class medical imaging; Synthetic radiology datasets	High computational cost; Instability at large batch sizes
Few-Shot GAN	Adapts pre-trained models to new domains with limited data; Optimizes parameter efficiency	Rare disease datasets; low-sample medical imaging	Sensitive to training parameters; Potential overfitting

DCGAN has been applied to generate realistic medical images (Liu et al. 2022b), augment datasets for rare diseases, and create privacy-preserving synthetic data (Ali et al. 2024). Its ability to synthesize high-quality images facilitates the training of AI models in scenarios where real data is limited or sensitive. One notable application is the generation of synthetic fingerprints for biometric research, as demonstrated in the work of Striuk and Kondratenko (2021). They introduced Adaptive DCGAN (ADCGAN) to generate high-fidelity fingerprint images that accurately replicate the texture and structure of real fingerprints. Their model addresses privacy concerns by producing synthetic datasets for forensic and biometric security applications without compromising sensitive personal data. By fine-tuning DCGAN's architecture, they achieved highly realistic synthetic fingerprints, ensuring a robust dataset expansion for machine learning tasks. Similarly, Liu et al. (2021a) explored the role of normalization techniques in improving the training stability and robustness of convolutional networks, including DCGAN. They proposed Convolutional Normalization (ConvNorm), a method that optimizes the Lipschitz constant of convolutional layers, improving the training process and reducing sensitivity to noise. This enhancement enables DCGAN to generate higher-quality images while maintaining stability during adversarial training. The study highlights how ConvNorm can be integrated into DCGAN to enhance its performance in healthcare applications, such as generating noise-robust medical images for diagnostic training datasets. Furthermore, a comprehensive overview of DCGAN by Farajzadeh-Zanjani et al. (2022) in a broader context of GAN-based architectures emphasized its improvements in stabilizing GAN training and its success in producing high-resolution images, making it a vital component in medical imaging tasks. The study also noted DCGAN's contribution to addressing challenges such as vanishing gradients and mode collapse, which are common in traditional GANs. They also illustrated the DCGAN's relevance in generating synthetic medical images for applications, such as tumor detection and segmentation, which require high precision and realism.

3.1.2 Conditional GAN

Conditional GAN (CGAN) is an extension of the original GAN architecture that integrates conditional inputs into the generator and discriminator. This enhancement allows CGAN to produce data that aligns with specific conditions, such as generating images of a particular medical category or synthesizing patient-specific data. The generator takes a latent vector combined with a condition (such as a label or target feature) to produce data conditioned on the specified attribute. The discriminator evaluates the generated data's authenticity and whether it meets the given condition. This ability to generate data with controlled attributes in healthcare has made CGAN a valuable tool for synthetic data generation (Jung et al. 2023). Its applications include generating rare disease images for diagnostic models (Giuste et al. 2023), augmenting class-imbalanced datasets, and creating privacy-preserving synthetic records that retain statistical fidelity while protecting patient confidentiality. Additionally, CGAN has been used to simulate disease progression, offering insights into complex medical conditions such as Alzheimer's (Jung et al. 2023) and cancer (Liu et al. 2021b; Tavorara et al. 2019).

Several studies highlighted the CGAN's potential and addressed various challenges in its application. Cho and Yoon (2020) introduced Conditional Activation GAN (CAGAN), a variant of CGAN designed to improve training efficiency and simplify hyperparameter

tuning. By integrating multiple GANs with shared hidden layers, CAGAN streamlines the architecture and enhances training speed. They also proposed a mixed-batch training technique to stabilize learning by maintaining consistent real-to-generated data ratios. Although not healthcare-specific, these advancements make CAGAN highly suitable for generating large-scale synthetic datasets in healthcare, particularly for addressing imbalanced conditions. Similarly, Farajzadeh-Zanjani et al. (2022) emphasized CGAN's adaptability in healthcare applications, including generating tumor-specific images, high-resolution diagnostic images, and fair data that mitigates demographic biases in medical datasets. However, the practical implementation of CGAN in healthcare is hindered by computational costs and the need for large, diverse training datasets. Chakraborty et al. (2024) provided a comprehensive survey of GAN variants, underscoring CGAN's utility in generating condition-specific synthetic data for applications, such as balancing datasets and augmenting rare medical images. They also highlighted challenges such as mode collapse and the reliance on optimized loss functions to ensure reliable and accurate data generation. These studies collectively illustrate CGAN's potential and limitations, providing a foundation for further research into improving its application in healthcare.

3.1.3 Wasserstein GAN

Wasserstein GAN (WGAN) is designed to address key issues in training traditional GANs, such as mode collapse, training instability, and difficulty in convergence. Unlike the original GAN, which uses the Jensen–Shannon divergence to measure the distance between the real and generated data distributions, WGAN introduces the Earth Mover (EM) distance (Wasserstein distance) as a loss function. This metric provides smoother gradients and ensures a more stable training process. WGAN achieves better alignment between the real and synthetic data distributions by optimizing the EM distance, significantly reducing the risk of mode collapse. The improved stability of WGAN has made it a robust choice for generating high-quality and diverse synthetic data, particularly in medical applications where such properties are significant. In healthcare, WGAN has shown potential for handling data imbalance issues and improving the generation of synthetic samples for rare disease cases. By producing realistic and diverse data, WGAN supports training diagnostic models in settings with limited availability of annotated medical data. It has been employed for applications such as medical image synthesis (Hussain et al. 2022), augmentation for rare disease datasets, and privacy-preserving synthetic data generation, which retains the essential characteristics of real data while protecting patient confidentiality.

Recent studies have demonstrated the significant capabilities and benefits of WGAN in healthcare applications and beyond. Wang et al. (2019) proposed a WGAN-based synthetic minority over-sampling technique to tackle data imbalance in lung nodule classification using CT images. Their findings highlighted how WGAN synthesizes realistic samples for minority classes, effectively enhancing the performance of fine-grained classification models. Compared to traditional augmentation methods and GAN variants like Deep Convolutional GAN, WGAN achieved superior results by better approximating the distribution of minority class samples. Such applications improve classification accuracy and support the development of reliable diagnostic models for rare conditions. Additionally, Adler and Lunz (2018) extended the traditional WGAN framework by introducing Banach Wasserstein GAN (BWGAN), a generalization that allows the use of Banach spaces. This advancement

enables practitioners to select norms emphasizing specific features in generated data, such as edges or global structures, improving its adaptability in image generation tasks. While their work focused on datasets like CIFAR-10 and CelebA, the insights can be applied to medical image synthesis, where the feature-specific emphasis is critical for accurate diagnosis. Furthermore, a survey by Farajzadeh-Zanjani et al. (2022) emphasized the role of WGAN in addressing training stability and mode collapse issues inherent in traditional GANs. The survey underscored WGAN's ability to stabilize adversarial training using the Wasserstein distance, which allows for generating high-quality medical images, such as CT scans and MRI data while supporting the development of balanced datasets for training diagnostic models. These studies demonstrate the versatility and effectiveness of WGAN in healthcare applications, showcasing its ability to address several challenges in synthetic data generation.

3.1.4 CycleGAN

CycleGAN is a powerful GAN variant designed for unpaired image-to-image translation, allowing transformations between two distinct domains without requiring paired data for training. The architecture consists of two generators and two discriminators: the first generator translates images from the source domain to the target domain. In contrast, the second generator reconstructs them back to the source domain. This process relies on cycle consistency, ensuring that a translation followed by a reverse translation recovers the original image. The inclusion of cycle consistency loss, combined with adversarial loss, stabilizes training and enhances the quality of generated images (Jabbar et al. 2021). CycleGAN's ability to map complex transformations in an unsupervised manner makes it particularly effective for applications where paired datasets are scarce, such as in healthcare.

CycleGAN has been extensively applied to medical image analysis for domain adaptation, style transfer, and data augmentation tasks. Its ability to address domain shifts in datasets, such as variations in imaging modalities, staining techniques, or acquisition protocols, is invaluable for building robust diagnostic models. For instance, a study by Lakshmi et al. (2023) evaluated CycleGAN's performance in transforming images between domains, including generating monet-style paintings from real-world images, and demonstrated its effectiveness in scenarios with limited paired data. The study highlighted the importance of optimizers, such as SGD and ADAM, and architectural tuning in improving CycleGAN's performance across various applications, including healthcare. Similarly, Breen et al. (2021) explored its application in medical image generalization, focusing on improving domain generalizability for mitosis detection in histopathology images. By leveraging stain-to-stain translation, CycleGAN enhanced performance in scenarios with domain shifts caused by differences in staining protocols or imaging devices, addressing inter-laboratory variability essential for real-world deployment. Additionally, Jabbar et al. (2021) provided a comprehensive overview of GAN variants, emphasizing CycleGAN's ability to stabilize training through cycle consistency loss. The survey underscored how CycleGAN addresses mode collapse, enhances the realism of generated images, and is a preferred choice for domain adaptation and style transfer in healthcare while observing challenges like training stability and computational cost. These findings demonstrate CycleGAN's adaptability and potential for unpaired image-to-image translation tasks in healthcare.

3.1.5 StyleGAN

StyleGAN architecture was designed to generate high-quality images with exceptional detail and control. Developed by Karras et al. (2019), StyleGAN introduces significant innovations, including using a style-based generator architecture and progressive training for high-resolution image synthesis. The generator employs a mapping network that transforms input latent vectors into intermediate-style vectors, which modulate each convolutional layer, providing fine-grained control over image attributes. This unique design allows for disentangled representations of features such as texture, color, and structure in the generated images. Noise inputs at various layers further enhance detail and variability (Farajzadeh-Zanjani et al. 2022; Sauer et al. 2023). StyleGAN has found extensive applications in fields that require high-resolution and high-quality synthetic images.

In healthcare, StyleGAN's ability to generate high-resolution synthetic medical images has proven invaluable for training diagnostic models, augmenting small datasets, and simulating diverse conditions through latent space manipulations. For example, Fetty et al. (2020) demonstrated StyleGAN's effectiveness in generating synthetic CT and MRI scans of pelvic malignancies, where latent space manipulation enabled precise control over features such as imaging modality, sex, and slice position. This capability addresses challenges posed by limited and sensitive medical datasets, facilitating the robust training of AI models for radiological evaluations and treatment planning. Sauer et al. (2023) also introduced StyleGAN-T, a refined variant optimized for text-to-image synthesis, with significant implications for healthcare. The enhanced architecture incorporates style transfer and stochastic noise, enabling precise control over the synthesis of detailed medical datasets, such as organ-specific or disease-specific imagery. These advancements highlight its versatility in addressing real-world challenges like enhancing radiological datasets while safeguarding patient confidentiality. Furthermore, a comprehensive review by Farajzadeh-Zanjani et al. (2022) underscored StyleGAN's contributions to mitigating data scarcity by generating realistic, domain-specific medical images. The review also highlighted StyleGAN's role in advancing AI models for tasks such as tumor segmentation, disease detection, and medical image analysis while signifying challenges such as computational cost, training complexity, and achieving domain-specific generalization. Together, these studies illustrate StyleGAN's transformative potential for healthcare applications.

3.1.6 Auxiliary classifier GAN

Auxiliary Classifier GAN (ACGAN) enhances the generative process by integrating class information into both the generator and discriminator. Unlike standard GANs, which rely solely on adversarial training, ACGAN introduces an auxiliary classifier in the discriminator. This classifier predicts the class label of real and generated data, ensuring that generated samples are realistic and class-specific. The generator learns to produce data that satisfies both adversarial and classification losses, leading to more robust and class-distinct synthetic data (Chakraborty et al. 2024). ACGAN has gained popularity in healthcare for its ability to generate high-quality labeled data, essential for training supervised models in medical applications.

ACGAN has been employed to address the challenge of imbalanced datasets, where certain disease categories may be underrepresented. By generating synthetic data with specific

labels, ACGAN aids in balancing datasets and improving the performance of diagnostic models. Chakraborty et al. (2024) emphasized ACGAN's ability to create diverse, labeled synthetic data for applications like rare disease classification. Integrating class-specific auxiliary loss improves the model's capacity to generate data that aligns closely with the desired clinical characteristics. Additionally, Park and Shin (2024) proposed a novel generator architecture with auxiliary branches designed to enhance information flow and stabilize training in GANs. While not exclusively focused on ACGAN, their approach demonstrates the importance of auxiliary components in improving the performance of generative models. By combining residual blocks with auxiliary branches, their model achieved superior results in terms of Inception Score (IS) and Frechet Inception Distance (FID) across multiple datasets, indicating the potential for similar architectural advancements in ACGAN implementations. These findings illustrate ACGAN's versatility and efficacy in generating high-quality, class-specific synthetic data for healthcare applications.

3.1.7 Progressive growing GANs

Progressive Growing GAN (ProGAN) is one of the advanced GAN architectures designed to generate high-resolution images while addressing training instability and mode collapse issues common in traditional GANs. Developed by Karras et al. (2017), ProGAN introduces a novel training methodology where both the generator and discriminator are progressively trained, starting with low-resolution images and gradually adding higher-resolution layers. This approach allows the models first to capture broad structures and progressively refine intricate details, significantly improving stability during training. ProGAN employs techniques like pixel normalization and mini-batch standard deviation to enhance convergence and diversity in the generated data. These advancements make ProGAN well-suited for applications requiring high-quality synthetic image generation, including medical imaging.

ProGAN has demonstrated immense potential for synthesizing high-resolution medical images, which is critical for diagnostics and research. Raner et al. (2025) explored the use of ProGAN to generate synthetic brain MRI, chest CT, and retina fundus images, achieving resolutions as high as 2048×2048 pixels. These images were successfully used to train classification models, yielding accuracy improvements of up to 91.47% in specific tasks. Their work highlights ProGAN's ability to address data scarcity and balance datasets for rare medical conditions, making it a valuable tool for advancing diagnostic model performance. Another study by Tetteh et al. (2024) applied ProGAN in hydrogeological modeling, showcasing its adaptability beyond standard medical applications. The study trained ProGAN to generate high-resolution geological models conditioned on hard data, such as hydraulic conductivity. These generated models were integrated with ensemble smoothers and data assimilation techniques, demonstrating enhanced predictive accuracy for complex geological structures. This work emphasizes ProGAN's capacity for handling diverse datasets and conditioning outputs on real-world data, a feature that can be directly applied to multimodal medical imaging. Furthermore, Golfe et al. (2023) introduced ProGleasonGAN, a conditional variant of ProGAN tailored for prostate cancer research. This framework synthesized histopathological patches conditioned on Gleason grades, a critical metric for prostate cancer diagnosis. The study achieved realistic image synthesis by embedding conditional information into ProGAN's architecture, validated through expert pathologist evaluations and the FID. The synthetic data was successfully used to augment imbalanced

datasets, improving the accuracy of classification models for prostate cancer grading. This study underscores ProGAN's adaptability and effectiveness in tackling data imbalance and enhancing AI models in pathology. These findings collectively demonstrate the transformative role of ProGAN in generating high-resolution, realistic, and domain-specific synthetic medical images. Its unique progressive training methodology ensures stability and high-quality outputs, making it a cornerstone for applications in healthcare.

3.1.8 Attention/self-attention GAN

Attention/self-attention GANs (AttnGAN and SAGAN) are generative models that integrate attention mechanisms into the GAN architecture. By focusing on both global and local dependencies, these models enhance the quality of generated images, particularly in capturing fine-grained details and long-range dependencies. SAGAN, introduced by Zhang et al. (2019), incorporates a self-attention mechanism that dynamically assigns attention weights to regions within an image, enabling the model to understand complex patterns and relationships. AttnGAN, on the other hand, extends these capabilities to text-to-image synthesis, where it aligns text descriptions with corresponding visual features using deep attention mechanisms. These models are particularly effective in healthcare, where they can generate detailed, domain-specific synthetic data for applications such as medical imaging and diagnostics.

AttnGAN has been applied to synthesize realistic medical images from text descriptions, offering significant utility in generating synthetic datasets for underrepresented conditions. For instance, a study by Ma et al. (2018) demonstrated AttnGAN's application in translating textual information into detailed images, highlighting its potential for augmenting medical datasets with descriptive labels. The model effectively aligned visual features with text by employing instance-level attention, reducing mode collapse, and improving image quality. This capability has significant implications for healthcare scenarios, such as generating visual representations of textual radiology reports. Similarly, SAGAN has enhanced medical imaging tasks by capturing long-range dependencies in image data. In a study by Marcos et al. (2021), the integration of boosting attention fusion blocks within SAGAN improved the denoising of low-dose CT images. This approach preserved structural details and prevented over-smoothing, a common limitation in conventional denoising models. Using spatial and channel attention modules enabled the model to focus on critical image features, such as blood vessels and small lesions, enhancing diagnostic accuracy while maintaining computational efficiency. These findings underline the transformative potential of attention-based GANs in healthcare. By incorporating attention mechanisms, AttnGAN and SAGAN provide tools for generating high-quality synthetic medical images, improving diagnostic capabilities, and addressing data scarcity and variability challenges.

3.1.9 BigGAN

BigGAN, introduced by Brock et al. (2019), designed to generate high-resolution images with exceptional quality. It employs architectural innovations like self-attention mechanisms, spectral normalization, and class-conditional batch normalization, which enable it to stabilize training, reduce mode collapse, and achieve higher diversity in generated images. The model operates with an expanded latent space and larger batch sizes, further enhancing

its ability to produce diverse and realistic outputs. These features make BigGAN particularly well-suited for applications requiring high-quality visual synthesis.

In healthcare, BigGAN's capacity for generating high-fidelity images has proven instrumental in advancing medical imaging tasks, especially in data augmentation and synthetic image generation for underrepresented conditions. For instance, Quaicoo et al. (2024) explored Triple-BigGAN, an extension of BigGAN, for healthcare-specific image detection tasks. By incorporating a co-trained classifier, the model enhanced its ability to generate images tailored to diagnostic needs. The study demonstrated BigGAN's adaptability within computational environments, such as Google Colab and Vertex AI, and evaluated its performance using metrics such as FID and IS. This highlighted BigGAN's potential to support diverse healthcare applications, including the generation of synthetic radiological datasets for diagnostic AI. Similarly, Sato and Iiduka (2023) investigated critical factors affecting BigGAN's training, such as batch size and learning rates. Their findings revealed the existence of an optimal batch size for achieving low FID scores, which is essential for producing realistic medical images. The study showcased improved stability and efficiency by optimizing training conditions, further validating BigGAN's suitability for large-scale medical data synthesis. These findings underscore BigGAN's ability to generate diverse, high-resolution synthetic medical images and its transformative potential to address data scarcity challenges and enhance diagnostic AI in healthcare.

3.1.10 Few-Shot GAN

Few-Shot GAN (FSGAN) addresses the limitations of traditional GANs in generating high-quality images with limited training data. Unlike conventional GANs that require large datasets, FSGAN adapts pre-trained GANs to new domains with a small number of samples (5–100 images). This is achieved by constraining the optimization process to highly expressive parameter spaces through Singular Value Decomposition (SVD) of pre-trained weights, freezing singular vectors, and adapting singular values (Robb et al. 2020; Phaphuangwittayakul et al. 2021). This novel approach ensures diverse and realistic image synthesis without overfitting, making FSGAN ideal for applications in healthcare, where labeled data is often scarce.

FSGAN has demonstrated significant potential in generating synthetic datasets for rare diseases or underrepresented conditions. For instance, Robb et al. (2020) introduced FSGAN to adapt pre-trained StyleGAN2 models for domains with limited data, such as toddler images from the CelebA dataset. This approach was validated on few-shot settings, highlighting improvements in visual quality compared to other GAN adaptation methods. The study emphasized FSGAN's capacity for smooth latent space interpolation and output diversity, ensuring that the model does not merely memorize training samples but generates novel, high-quality samples. Another study by Phaphuangwittayakul et al. (2021) proposed a Fast Adaptive Meta-Learning (FAML) framework that combines FSGAN with an encoder network for few-shot image generation. This approach focuses on meta-learning to achieve tenfold faster convergence and higher efficiency with fewer trainable parameters than baseline models. Applied to datasets like VGG-Faces and miniImageNet, FAML demonstrated improvements in FID scores and diversity metrics, underlining its suitability for high-dimensional medical datasets and its ability to generate synthetic training data for diagnostic applications. Further advancements were highlighted by Qirui et al. (2021),

who integrated FSGAN with a pruning optimization method using Adam. Their approach significantly reduced model complexity while maintaining accuracy, allowing for the synthesis of diverse medical images for classification tasks. The study applied this technique to datasets such as ISIC-2018 for skin lesion analysis, demonstrating FSGAN's robustness in augmenting datasets for healthcare research. These findings emphasize Few-Shot GAN's transformative impact on healthcare, providing a reliable method to generate high-quality synthetic data with limited training samples. Beyond few-shot adaptation, transfer learning strategies have also been leveraged to improve generative model training in healthcare. For instance, Schurer et al. (2025) fine-tuned pre-trained GANs to synthesize EHR for chronic kidney disease, demonstrating that transfer learning can mitigate data scarcity while maintaining clinical plausibility. Similarly, Gao and Cui (2020) highlighted how deep transfer learning can reduce disparities arising from biomedical data inequality, emphasizing its role in equitable model development.

3.1.11 Summary and open issues

GANs share common strengths, such as (i) high-fidelity sample generation with strong perceptual realism, (ii) flexibility for unpaired cross-modality translation, and (iii) broad adoption across imaging and structured data tasks. However, they also exhibit recurring limitations, such as (i) training instability and mode collapse, (ii) a lack of clinically aligned evaluation standards, and (iii) susceptibility to privacy leakage if adversarial audits are applied. Importantly, while visual fidelity metrics such as FID and SSIM are widely reported, they do not always align with downstream diagnostic performance or clinical trustworthiness. A consolidated overview of these common patterns is presented in Table 3. Future directions include hybrid GAN-diffusion approaches to balance fidelity and diver-

Table 3 GAN family—common strengths, use-cases, limitations, and future directions

Aspect	Strengths	Healthcare use-cases	Limitations and challenges	Future directions
Data fidelity	High-resolution synthesis, realistic textures	Radiology augmentation; histopathology patches	Mode collapse; anatomical inconsistencies	Hybrid GAN-diffusion models for higher stability
Data scarcity	Augments rare cases; conditional GANs improve minority classes	Rare tumor and lesion generation	Overfitting on very small datasets	Semi-supervised and transfer learning
Unpaired translation	Supports unpaired modality shifts	MRI to CT, CT to PET, cross-scanner harmonization	Risk of hallucination	Anatomical priors, domain adaptation
Privacy and sharing	Surrogate datasets without sharing raw data	Multi-institutional imaging; EHR surrogates	Membership/attribute inference risk	DP-SGD, federated GANs
Training stability	WGAN-GP reduces collapse; StyleGAN improves disentanglement	Stable high-resolution imaging	Still unstable in heterogeneous data	Alternative losses, spectral normalization
Evaluation standards	Benchmarks with FID, IS, PSNR, SSIM	Radiology and dermatology	Metrics not optimal for clinical utility	Task-aware, clinician-in-the-loop evaluation
Deployment	Flexible augmentation and harmonization	Low-dose CT denoising, reconstruction	Limited trust in regulatory settings	Transparent audit pipelines, FL/DP integration

sity, differentially private GAN training to mitigate leakage risks, and federated/distributed architectures that enable multi-center deployment without sharing raw data.

3.2 Variational autoencoder-based generative models

Variational autoencoders (VAEs) are probabilistic generative models that learn latent variable representations through an encoder-decoder framework, trained by maximizing evidence lower bound (ELBO). Their strength derives from structured, interpretable latent spaces that support uncertainty estimation, anomaly detection, and data completion. In healthcare, VAEs are particularly effective for structured EHR generation, time-series modeling, and signal denoising, while also serving as foundations for hybrid models that integrate diffusion or transformer decoders. Numerous variants, including β -VAE, Conditional VAE, InfoVAE, and Vector-Quantized VAE, extend this baseline to improve disentanglement, conditional control, or fidelity. Detailed variant-level discussions, including Table 4, follow in the subsections below.

Table 4 Comparison of variational autoencoder variants in healthcare applications

Variant	Key features	Supported data types	Limitations
Variational Autoencoders	Probabilistic latent space modeling; Smooth and diverse data generation	Medical imaging; Omics data; Structured EHR	Prone to over-regularization; Potential blurriness in generated images
Convolutional VAEs	Uses convolutional layers to capture spatial hierarchies; Enhances feature extraction in images	Medical imaging (X-rays, MRI, CT); Histopathology	Computationally expensive; Sensitive to hyperparameter tuning
Sparse VAEs	Enforces sparsity constraints in the latent space for feature selection	Neuroimaging; Multimodal healthcare data; Disease clustering	Risk of information loss if sparsity is too high
Stacked VAEs	Multi-layer hierarchical encoding improves feature extraction and generative quality	Drug discovery; Medical imaging; EHR synthesis	Computationally expensive; Risk of overfitting with deep stacking
Recurrent VAEs	Captures temporal dependencies in sequential data; Handles irregular time-series data	ECG, EEG, EHR time-series; Disease progression	Difficult to train on long sequences; Risk of vanishing gradients
Contractive VAEs	Introduces contractive penalty for robust latent representations; Enhances model stability	Structured healthcare data; EHR synthesis	Increased computational overhead; Requires fine-tuning of regularization strength
Vector Quantized VAEs	Uses discrete latent variables for structured feature encoding; Overcomes blurriness in VAEs	Medical imaging; Pathology slides; Genomic sequence modeling	Training instability; High memory requirements
Disentangled VAEs	Enforces feature disentanglement in the latent space for interpretable representations	Disease phenotype modeling; Medical imaging; Genomics	Higher values of β may reduce reconstruction quality
Variational Graph Autoencoders	Learns structured latent representations of graph-based data; Effective in relational datasets	Patient networks; Omics data; EHR; Disease progression analysis	Requires graph-based input structures; Limited generalization to non-graph data

3.2.1 Variational autoencoders

Standard VAEs extend conventional autoencoders by incorporating Bayesian inference. Unlike standard autoencoders that focus on deterministic feature extraction, VAEs learn a probabilistic latent space distribution, enabling them to generate diverse and high-quality synthetic data. This probabilistic approach ensures that the latent representations maintain smoothness and variability, making VAEs particularly useful for generating new, yet realistic, medical data (Wei and Mahmood 2020). VAEs are widely used in healthcare applications such as medical image reconstruction, disease diagnosis, EHR synthesis, and multi-omics data analysis. Their ability to model complex distributions while preserving statistical properties makes them a powerful tool for addressing data scarcity in medical AI.

Singh and Ogunfunmi (2021) explored the role of VAEs in bio-signal processing, demonstrating their ability to generate synthetic electrocardiogram (ECG), electromyogram (EMG), and electroencephalogram (EEG) signals. Their findings indicated that VAEs not only improved data augmentation but also enabled source separation in noisy environments, enhancing the quality of biomedical signal analysis. Similarly, Wei and Mahmood (2020) reviewed VAE applications in biomedical informatics, emphasizing their contributions to molecular design, protein structure analysis, and large-scale biological sequence modeling. Their research highlighted how VAEs improve intra-class variations within datasets, allowing for better representation learning and more effective downstream AI applications. Additionally, Akkem et al. (2024) examined the use of VAEs in smart farming, particularly in generating synthetic crop recommendation data. Although not directly related to healthcare, their work underscored VAEs' capability in structured data synthesis, an aspect critical for medical applications such as synthetic patient data generation for privacy-preserving research. These studies collectively demonstrate that VAEs provide a robust and scalable approach to synthetic data generation in healthcare. By leveraging probabilistic modeling, VAEs enhance data diversity while maintaining fundamental structural and statistical integrity, making them essential for various biomedical applications.

3.2.2 Convolutional variational autoencoders

Convolutional variational autoencoders (CVAEs) incorporate convolutional layers in both the encoder and decoder, making them particularly suited for analyzing and generating medical image data. CVAEs capture spatial hierarchies more effectively than fully connected standard VAEs, preserving local structures and improving the quality of synthesized medical images. This capability is imperative in healthcare applications such as disease diagnosis, medical image reconstruction, and anomaly detection, where fine-grained visual details play a significant role in decision making (Biswal et al. 2021). By leveraging convolutional networks, CVAEs enhance feature extraction and facilitate more accurate latent space representations, enabling them to generate high-fidelity synthetic images for training AI models while preserving data privacy.

Wan et al. (2019) applied a deep CNN-based VAE for semisupervised representation learning in the screening of infants for biliary atresia, a serious pediatric liver disease. Their approach involved collecting stool images using smartphone cameras and training a CVAE to differentiate between normal and acholic stools, achieving an accuracy of 92.16%. This study demonstrated that CVAEs can provide an effective alternative to traditional stool

color card screening, improving early detection and clinical outcomes. Similarly, Biswal et al. (2021) developed the EHR Variational Autoencoder (EVA), a CVAE-based model for synthesizing realistic longitudinal EHRs. Their experiments on a dataset of over 250,000 patients confirmed that synthetic EHRs generated by EVA closely resembled real patient records, addressing data privacy concerns while supporting disease-specific research and predictive modeling. These studies highlight the versatility of CVAEs in both medical imaging and structured healthcare data generation, reinforcing their role in improving disease diagnosis, data augmentation, and privacy-preserving AI applications in healthcare.

3.2.3 Sparse variational autoencoders

Sparse variational autoencoders (Sparse VAEs) enforces sparsity constraints on the latent space, making them particularly effective for learning meaningful and interpretable latent representations. Conventional VAEs typically assume a Gaussian prior for the latent variables, whereas Sparse VAEs incorporate Laplacian priors that promote sparsity. This structured representation allows the model to efficiently encode high-dimensional medical data, enabling applications in anomaly detection, feature selection, and multimodal data analysis. Sparse VAEs have been particularly useful in handling heterogeneous healthcare datasets, where reducing redundant or noisy information is critical for accurate decision making (Geadah et al. 2018; Antelmi et al. 2019; Sun et al. 2018). By enhancing the interpretability of the latent space, these models contribute to improved performance in biomedical applications such as neuroimaging analysis, patient phenotype clustering, and synthetic data generation.

Several studies highlighted the performance of Sparse VAEs for various tasks, such as Goubeaud et al. (2021) proposed a Sparse VAE model to generate synthetic time-series data for classification tasks, demonstrating that the model improved accuracy and F1-score across multiple datasets from the UCR Time Series Classification Archive. Their approach augmented small training sets by generating diverse yet realistic time-series patterns, addressing data scarcity issues in clinical time-series analysis. Similarly, Geadah et al. (2018) introduced a Sparse-Coding Variational Autoencoder (SVAE) that incorporates an overcomplete latent representation and a sparse prior, which enhances feature extraction for medical imaging applications. Their study showed that SVAE-based models achieve superior log-likelihood and capture biologically meaningful representations in neuroscience data. Antelmi et al. (2019) developed a Sparse Multi-Channel Variational Autoencoder (SMCVAE) for analyzing heterogeneous medical data, including clinical scores and neuroimaging features, effectively disentangling disease-related factors from demographic variations. Additionally, Sun et al. (2018) leveraged a Sparse VAE for anomaly detection in healthcare applications, including network intrusion detection and medical image anomaly screening, showing its effectiveness in identifying rare pathological cases. These studies collectively demonstrate that Sparse VAEs provide a powerful and interpretable framework for healthcare applications, enhancing both classification performance and data efficiency while addressing the challenges of high-dimensional medical datasets.

3.2.4 Stacked variational autoencoders

Stacked variational autoencoders (Stacked VAEs) leverage a hierarchical architecture where multiple VAEs are trained sequentially to refine latent representations. This design enhances the model's ability to learn complex distributions and generate higher-quality synthetic data. Stacked VAEs are particularly useful in applications that require multi-stage feature extraction, such as medical imaging, drug discovery, and structured data synthesis. By progressively refining the latent space, these models ensure that critical features are preserved while eliminating redundant information, making them valuable for healthcare applications where interpretability and feature disentanglement are of utmost importance.

Tibebu et al. (2022) proposed a Stacked Conditional Variational Autoencoder (Stack-CVAE) for text-to-image synthesis, demonstrating its effectiveness in generating high-resolution images from textual descriptions. Their model used a two-stage approach: first, a CVAE generated a rough sketch of the image, and then a conditional GAN refined the output into a high-resolution image. This method significantly improved image quality and diversity while maintaining conditioning consistency, which is essential for applications like medical imaging synthesis. Similarly, Kim et al. (2022) developed a Stacked Augmented Conditional VAE for de novo drug discovery, integrating reinforcement learning to optimize molecular generation based on binding affinity predictions. Their model outperformed conventional generative approaches by producing valid and unique chemical compounds with enhanced drug-likeness and synthesizability. These studies illustrate the potential of Stacked VAEs in generating complex and domain-specific synthetic data for healthcare applications, offering improvements in both quality and interpretability.

3.2.5 Recurrent variational autoencoders

Recurrent variational autoencoders (RVAEs) enhance traditional VAEs by incorporating recurrent neural networks (RNNs) into their encoder and decoder structures, making them particularly effective for modeling sequential and time-series data. RVAEs can capture long-term dependencies and temporal variations, making them useful in healthcare applications where data evolves over time, such as patient monitoring, disease progression modeling, and synthetic time-series data generation (Jeon and Seo 2023). These models often integrate mechanisms such as attention layers or adversarial training to enhance the quality and interpretability of generated sequences. By learning the probabilistic latent space of sequential data, RVAEs provide robust solutions for handling missing values, irregular sampling, and complex temporal dependencies in medical datasets.

In a separate study, Sun et al. (2021) developed LongGAN, a recurrent autoencoder-based approach combined with a Wasserstein GAN to generate synthetic longitudinal electronic health records (EHRs). Their results showed that predictive models trained on synthetic EHRs generated by LongGAN achieved performance comparable to real-world data while maintaining patient privacy. Jeon and Seo (2023) introduced an attention-based Variational Recurrent Autoencoder (AVRAE) to generate synthetic time-series data for industrial control systems (ICS), extending the evidence lower bound formulation to sequential data. Their model demonstrated the ability to learn short- and long-term dependencies using an attention mechanism, significantly improving the quality of synthetic time-series data. These studies highlight the potential of RVAEs in generating realistic and privacy-preserv-

ing synthetic healthcare data, supporting applications such as disease prediction, treatment recommendation, and personalized medicine.

3.2.6 Contractive variational autoencoders

Contractive variational autoencoders (Contractive VAEs) enhance traditional VAEs by introducing a contractive penalty to the loss function, which encourages robustness against small input perturbations. This regularization term, typically based on the Frobenius norm of the Jacobian matrix of the encoder's output with respect to its input, forces the model to learn invariant latent representations. This property is particularly useful in healthcare applications where small variations in medical data should not lead to drastically different representations. Contractive VAEs are effective for structured data synthesis, anomaly detection, and privacy-preserving data generation in clinical settings, making them a valuable tool for handling high-dimensional medical datasets while ensuring stability and interoperability.

Fallahian et al. (2024) introduced a Contractive Autoencoder (CAE)-based framework to improve tabular data generation using Conditional Tabular GANs (CTGANs). Their approach leveraged a pre-trained CAE to generate context-aware noise, which replaced the standard random noise input used in GAN-based tabular data synthesis. Experiments conducted on multiple benchmark datasets demonstrated that incorporating contractive representations enhanced the fidelity and semantic consistency of synthetic data. The study also showed that CAE-enhanced CTGANs improved machine learning model performance in downstream predictive tasks, reinforcing the effectiveness of contractive constraints in preserving structural properties in synthetic datasets. These findings highlight the potential of Contractive VAEs in stabilizing generative models, improving synthetic data realism, and ensuring privacy in healthcare applications such as patient record synthesis and structured medical data augmentation.

3.2.7 Vector quantized variational autoencoders

Vector Quantized Variational Autoencoders (VQ-VAEs) incorporate discrete latent representations into traditional VAEs through vector quantization. Standard VAEs assume a continuous latent space, whereas VQ-VAEs map encoded inputs to a finite set of learned embedding vectors. This discrete representation improves the model's ability to capture complex data structures while reducing mode collapse, making VQ-VAEs particularly useful for healthcare applications involving medical image recovery, structured EHR synthesis, and anomaly detection (Tarhouni et al. 2023; Gao et al. 2024). The quantization mechanism allows better interpretability and robustness against memorization, ensuring privacy preservation in sensitive medical data applications.

Lee et al. (2024) proposed CodeAR, a medical time series generative model that applies autoregressive modeling on discrete tokens obtained from a VQ-VAE. Their study demonstrated that CodeAR preserves patient privacy while outperforming GAN-based models in the distribution modeling of electronic health records (EHRs). Similarly, Shwetha et al. (2024) utilized VQ-VAE for data augmentation of Gram-stain images, demonstrating that synthetic images generated with this approach significantly improved classification accuracy in bacterial infection detection. Tarhouni et al. (2023) introduced ReVQ-VAE, a VQ-VAE-based approach for recovering corrupted COVID-19 chest X-ray images, showing

improved SSIM scores compared to conventional recovery techniques. Gao et al. (2024) further extended VQ-VAE by integrating Support Vector Data Description (SVDD) to improve anomaly detection in lung CT images, achieving superior area under the ROC curve scores compared to traditional VAEs. These studies collectively highlight the potential of VQ-VAEs in healthcare, offering robust solutions for synthetic data generation, medical image enhancement, and privacy-preserving AI applications.

3.2.8 Disentangled variational autoencoders

Disentangled Variational Autoencoders (β -VAE) extend traditional VAEs by incorporating an additional hyperparameter, β , which regulates the trade-off between reconstruction quality and latent space regularization. By increasing β , the model enforces a more structured and independent latent representation, promoting the disentanglement of features within the data. This allows β -VAE to learn representations in which individual latent dimensions correspond to specific factors of variation, making it useful for interpretable generative modeling (Mathieu et al. 2019). In healthcare, β -VAEs have been widely applied in medical imaging, genomics, and disease subtype discovery, enabling structured feature extraction from complex datasets such as MRI scans, pathology images, and transcriptomics data.

Mathieu et al. (2019) introduced a decomposition framework for β -VAE, emphasizing its ability to learn independent latent variables by controlling the level of overlap in the latent space. Their theoretical analysis demonstrated that proper adjustments in the prior distribution could improve the disentanglement of latent representations while maintaining reconstruction fidelity. This study provided insights into optimizing β -VAE for tasks requiring interpretable latent representations, such as disease phenotype modeling. Additionally, Eltager et al. (2023) benchmarked different VAE models, including β -VAE, on cancer transcriptomics data to evaluate their ability to capture biologically relevant variations. Their study highlighted that β -VAE, when tuned correctly, improve clustering performance and survival prediction in cancer datasets, demonstrating its capability to extract meaningful latent features from high-dimensional biological data. These studies underscore the potential of β -VAE in healthcare applications, particularly in tasks requiring structured and interpretable latent representations. By leveraging disentanglement, β -VAEs facilitate improved disease classification, biomarker discovery, and personalized medicine through more transparent and generalizable deep learning models.

3.2.9 Variational graph autoencoders

Variational graph autoencoders (VGAEs) are designed to learn latent representations of structured graph data. Unlike traditional autoencoders, which operate on tabular or image data, VGAEs utilize graph structures to capture complex dependencies among data points. These models are particularly effective in healthcare applications such as patient trajectory modeling, synthetic data generation, and cell clustering, where relationships between entities (such as genes, patients, or medical events) play a vital role in analysis (Nikolentzos et al. 2023). VGAEs integrate graph convolutional networks (GCNs) for feature extraction, followed by an encoding-decoding process that learns and reconstructs the underlying graph

structure. VGAEs enhance GAEs by incorporating latent variable models, making them highly suitable for healthcare applications requiring robust data representation.

A study by Lin et al. (2023) introduced a Gaussian Mixture Model-based Variational Graph Autoencoder (scGMM-VGAE) for clustering single-cell RNA sequencing (scRNA-seq) data. Their approach combined statistical clustering with deep learning, significantly improving cell clustering performance compared to traditional methods. The model leveraged a cell-cell graph adjacency matrix and a gene feature matrix to generate latent representations, which were then clustered using a Gaussian Mixture Model (GMM). The integration of probabilistic modeling into GAEs enabled the detection of rare tumor cells, which is a critical aspect of precision medicine. Similarly, Nikolentzos et al. (2023) applied Variational Graph Autoencoders (VGAEs) to generate synthetic EHRs, addressing the challenge of patient privacy in data sharing across healthcare organizations. Their model produced realistic patient trajectories by representing clinical events as graph-based sequences, preserving both statistical properties and privacy. The study demonstrated that VGAE-generated EHRs maintain clinical realism while complying with privacy regulations such as HIPAA and GDPR, making them a promising solution for privacy-preserving AI in healthcare. These studies highlight the potential of Graph Autoencoders in healthcare applications, particularly in medical data synthesis, patient trajectory analysis, and omics data clustering. By leveraging graph-based learning, GAEs enable more accurate, interpretable, and scalable representations of complex biomedical relationships, contributing to advancements in precision medicine and medical AI.

3.2.10 Summary and open issues

VAEs provide stable training, probabilistic reasoning, and interpretable latent spaces, making them valuable for EHR synthesis, ICU time-series modeling, and phenotype discovery. However, they are often outperformed by GANs and diffusion models in high-fidelity image synthesis, with reconstructions suffering from blurriness due to reliance on pixel-level loss functions. Moreover, posterior collapse and the fidelity-disentanglement trade-off limit downstream clinical utility. These shared properties, including strengths, use-cases, limitations, and prospective directions, are summarized in Table 5. Future advances may come from hybrid approaches (such as VAE with diffusion decoders), domain-specific priors for clinical interpretability, and federated VAE architectures that preserve privacy while enabling multi-institutional training.

3.3 Transformer-based generative models

Transformers are sequence-to-sequence architectures that rely on self-attention mechanisms to capture long-range dependencies. Their flexibility allows them to model text, images, signals, or multimodal combinations, making them highly relevant for healthcare applications that span clinical note synthesis, multimodal report generation, and structured EHR sequence modeling. Autoregressive Transformers (GPT-style), Masked Transformers (BERT-style), and hybrid architectures (Vision Transformers, multimodal VQ-VAE with Transformers) have been adopted to generate synthetic clinical narratives, align imaging with text reports, and integrate heterogeneous data sources. The subsections below, including Table 6, provide detailed variant-level discussions.

Table 5 VAE family—common strengths, use-cases, limitations, and future directions

Aspect	Strengths	Healthcare use-cases	Limitations and challenges	Future directions
Structured generation	Probabilistic latent variables; uncertainty estimation	Synthetic EHR and tabular data; ICU time-series	Mixed-type data integration remains difficult	Domain-specific priors; unified heterogeneous encoders
Interpretability	Latent disentanglement enables factor analysis	Phenotype discovery; biomarker association; counterfactuals	Posterior collapse; trade-off with fidelity	Improved objectives ($\beta - VAE$, InfoVAE); KL annealing
Stable training	Convex-like ELBO objective ensures convergence	Large-scale EHR synthesis; anomaly detection	Limited expressiveness vs GAN/diffusion	Hybrid decoders (VAE-GAN, VAE-diffusion)
Signal modeling	Encoder-decoder works well with sequential inputs	ECG/EEG time-series synthesis and denoising	Temporal dependencies harder to capture	Sequential VAEs; transformer-VAE hybrids
Image reconstruction	Well-calibrated density estimation	Medical image denoising; anomaly detection	Blurry reconstructions; low perceptual fidelity	Perceptual loss; VQ-VAE; diffusion-based decoders
Privacy	Latent representation obfuscates patient-level identifiers	Privacy-preserving EHR generation	Latent leakage via inversion possible	Federated/ distributed VAEs with DP guarantees

Table 6 Comparison of transformer variants in healthcare applications

Variant	Key features	Supported data types	Limitations
Generative Pre-trained Transformers variants	Large-scale transformer models with autoregressive text generation; Excels in medical text summarization, question answering, and EHR synthesis	Medical reports; Clinical text; Structured/unstructured EHR data	Requires extensive fine-tuning; Prone to hallucination and bias
Vision Transformer	Self-attention mechanism for medical image processing; Effective in classification, segmentation, and anomaly detection	MRI; CT scans; Histopathology slides; X-rays; ECGs	Computationally intensive; Requires large datasets for optimal performance
Bidirectional and Auto-Regressive Transformer	Combines bidirectional encoding with autoregressive decoding; Useful for clinical text generation, summarization, and automated reporting	Radiology reports; EHR narratives; Structured text	Requires domain-specific adaptation for medical terminology; High training cost
Masked Generative Image Transformer	Parallel decoding mechanism for image synthesis; Efficient in medical image augmentation and anomaly detection	MRI; CT; Ultrasound; Histopathology images	Limited real-world validation; Requires labeled datasets for medical adaptation
Masked Generative Video Transformer	Generates temporally consistent video frames using masked token modeling; Useful for medical video reconstruction and anomaly detection	Endoscopy; Ultrasound; Surgical videos; Patient monitoring	Computationally expensive; Requires extensive labeled video datasets
DALL-E, DALL-E 2, DALL-E 3	Text-to-image synthesis; Generates synthetic medical images for training, education, and research	Radiology images; Anatomical diagrams; Pathology slides	Ethical concerns over synthetic data validity; Potential biases in generated images
Perceiver and Perceiver IO	Latent space-based transformer for multimodal data integration; Excels in heterogeneous medical data fusion	Multimodal patient records; Genomic data; Medical imaging	Requires complex pre-processing for optimal performance; Interpretability challenges

3.3.1 Generative pre-trained transformers

Generative Pre-Trained Transformers (GPT) have revolutionized natural language processing (NLP) and AI-driven text generation, enabling sophisticated applications in various domains, including healthcare. The GPT architecture is based on self-attention mechanisms and transformer models, allowing it to understand and generate human-like text. These models are trained on large-scale datasets and fine-tuned for specific tasks, such as clinical text generation, medical summarization, predictive analytics, and personalized health assistants. Recent GPT variants, such as GPT-3, GPT-4, and GPT-NeoX-20B (Black et al. 2022), have demonstrated enhanced reasoning, contextual understanding, and improved accuracy in domain-specific applications. GPT models are leveraged for automated medical documentation, clinical decision support, and AI-assisted diagnosis in healthcare settings. They can synthesize EHRs, generate medical reports, assist in drug discovery, and enhance patient engagement through AI-driven virtual assistants.

GPTs, however, have demonstrated significant potential in synthetic data generation for healthcare applications, particularly in addressing data scarcity, enhancing feature extraction, and improving data quality. BioGPT, proposed by Luo et al. (2022), is a domain-specific GPT model pre-trained on 15 million PubMed abstracts for biomedical text generation and mining. It outperforms previous models in biomedical NLP tasks such as relation extraction, question answering, and document classification, achieving a record-breaking F1-score of 44.98% on BC5CDR and 78.2% accuracy on PubMedQA. Similarly, Fahim Sufi's study (Sufi 2024) explores GPT's ability to synthesize structured patient discharge summaries, generating over 70 synthetic records with 20 distinct fields, including severity and hospital readmission probabilities. The research underscores GPT's effectiveness in generating labeled medical data, demonstrating an average precision, recall, and F1-score of 0.95, 0.97, and 0.96, respectively. Beyond data generation, GPT models are also being leveraged for medical decision support and forensic analysis of synthetic datasets. Atallah et al. (2023) discuss GPT-3 and GPT-4's impact on clinical decision-making, medical documentation, and predictive analytics, highlighting its integration with electronic medical records (EMRs) to automate discharge summaries and optimize patient triage. Meanwhile, Busker et al. (2025) analyze the quality and biases of GPT-generated synthetic data, performing granularity-based similarity assessments to evaluate its statistical properties. Their study suggests that well-designed prompts and targeted model training can mitigate biases and improve the usability of synthetic healthcare data. Collectively, these studies emphasize the growing role of GPT in healthcare AI, particularly in generating high-quality synthetic datasets that can enhance model training, privacy preservation, and decision support systems. However, challenges such as bias, ethical concerns, and regulatory compliance remain significant factors for their adoption in clinical settings.

3.3.2 Vision transformer

Vision Transformers (ViT) are a powerful alternative to CNNs in medical imaging and diagnostic applications. Unlike CNNs, which rely on local feature extraction through convolutions, ViTs leverage self-attention mechanisms to capture global dependencies across image regions, making them particularly effective for complex medical imaging tasks. The architecture of ViTs divides an image into patches, processes them as sequences, and applies

multi-head self-attention to learn spatial relationships. This enables superior performance in classification, segmentation, and anomaly detection for medical imaging modalities such as MRI, CT scans, histopathology slides, and ECGs. ViTs also support transfer learning, where models pre-trained on large datasets can be fine-tuned for specific medical tasks, improving efficiency in low-data scenarios (Vaid et al. 2023).

ViTs have been increasingly integrated into healthcare AI, such as ResViT, introduced by Dalmaz et al. (2022), is a Residual Vision Transformer designed for multimodal medical image synthesis, combining the strengths of CNNs and ViT. The model employs Aggregated Residual Transformer (ART) blocks, which synergistically fuse local CNN-based precision with global transformer-based contextual understanding. ResViT demonstrated superior performance in generating missing MRI sequences and synthesizing CT images from MRI scans, outperforming CNN-based synthesis models in both qualitative and quantitative assessments. A key innovation of ResViT is its weight-sharing strategy, which reduces computational costs while preserving model capacity, making it more feasible for real-world clinical applications. In another study, MedSynth, proposed by Kanagavelu et al. (2024), applies a Vision Transformer-based dataset condensation approach to enhance privacy-preserving healthcare data sharing. Instead of directly exchanging raw medical images, MedSynth utilizes an attention-based generator and ViT to produce representative synthetic datasets that encapsulate key medical features. This method significantly reduces privacy risks while maintaining data utility for downstream tasks such as disease classification. Extensive experiments on the ISIC 2019 (skin cancer) and Alzheimer's MRI datasets demonstrated that MedSynth outperforms traditional dataset condensation techniques while resisting Membership Inference Attacks (MIA), making it a promising approach for federated medical AI. These studies highlight the evolving role of Vision Transformers in medical AI, offering novel solutions for image synthesis, privacy-preserving data sharing, and clinical decision support. However, challenges such as computational complexity, the need for extensive labeled datasets, and ethical considerations in AI-driven diagnostics remain areas for further research and refinement.

3.3.3 Bidirectional and auto-regressive transformer

Bidirectional and auto-regressive transformers (BART) combine the advantages of bidirectional encoding (BERT; Sun et al. 2019) and auto-regressive decoding (GPT), making them highly effective for text generation, summarization, and sequence-to-sequence learning tasks. BART is a denoising autoencoder trained to reconstruct corrupted text, which enhances its robustness in handling medical texts, clinical summarization, and medical report generation. The model is particularly valuable in medical documentation, patient records synthesis, and automated report generation, as it efficiently captures long-range dependencies and contextual information (Guan et al. 2021). BART-based models have been used in healthcare for automated clinical note generation, medical dialogue systems, and predictive text generation for radiology and pathology reports. The model's bidirectional encoding allows it to understand complete contexts, while the auto-regressive decoding ensures coherent and structured text output.

Zhou et al. (2023) proposed a lightweight parameterized model integrating BART with a pointer-generator network for improved news text generation, demonstrating efficient text synthesis with reduced parameters. While the study focused on journalism, the model's

ability to generate coherent and structured narratives makes it applicable to medical report automation and clinical text summarization. Another work by Yu et al. (2021) introduced BAT-Fill, a Bidirectional and Auto-Regressive Transformer (BAT) designed for image inpainting. While BAT-Fill shares architectural similarities with BART, both combining bidirectional encoding with autoregressive decoding, it is not the same model. BAT-Fill is a specialized adaptation for image inpainting tasks, whereas BART was originally developed for text generation. Their study highlights BART's versatility in handling structured and unstructured data, which is beneficial for medical image captioning, diagnostic interpretation, and structured clinical reporting. Additionally, Guan et al. (2021) explored BART's application in long text generation, particularly in improving coherence, discourse structure, and context understanding. Their findings suggest that BART-based models can significantly enhance the quality and readability of AI-generated medical literature, clinical narratives, and automated diagnostic summaries. These studies highlight BART's transformative role in healthcare, enabling efficient, context-aware text generation for medical documentation, report automation, and clinical dialogue systems. However, challenges such as fine-tuning domain-specific medical jargon, maintaining factual accuracy, and addressing biases in model outputs remain areas for further research and development.

3.3.4 Masked generative image transformers

Masked Generative Image Transformers (MaskGIT) are a powerful generative AI model that improves image synthesis by leveraging Bidirectional Transformers for parallel decoding. Unlike conventional autoregressive models that process images sequentially, MaskGIT predicts all image tokens simultaneously, refining them iteratively to generate high-resolution medical images. This architecture is particularly valuable in medical imaging applications, such as MRI and CT scan reconstruction, image inpainting, and anomaly detection (Chen et al. 2023). By utilizing a masked prediction approach similar to BERT's masked language modeling, MaskGIT ensures efficient training, reduced computational complexity, and enhanced generation quality.

MaskGIT has been applied in medical diagnostics, image augmentation for rare conditions, and improving the interpretability of medical imaging AI models. For instance, Chang et al. (2022) introduced MaskGIT, demonstrating its ability to generate high-fidelity images with significantly reduced inference time compared to autoregressive models. Their results highlighted that MaskGIT's parallel decoding mechanism improves synthesis speed by up to 48 times while maintaining superior image quality. Marimont et al. (2023) explored MIM-OOD, an unsupervised out-of-distribution (OOD) detection framework for medical images based on generative masked image modeling. This approach effectively identified anomalies in brain MRIs, improving accuracy and inference speed in detecting glioma lesions. Furthermore, Chen et al. (2023) extended masked image modeling (MIM) to 3D medical imaging analysis, demonstrating its advantages in medical image segmentation and self-supervised learning. Their findings showed that MIM-based models accelerate training convergence and improve the accuracy of 3D medical image reconstructions, making them a promising tool for MRI, CT, and ultrasound imaging. These studies collectively highlight MaskGIT's potential in medical imaging, offering faster, high-quality image generation and enhanced anomaly detection capabilities. However, challenges such as scalability to high-

dimensional medical images, the need for extensive labeled datasets, and real-world clinical validation remain key areas for further exploration.

3.3.5 Masked generative video transformers

Masked Generative Video Transformer (MAGVIT) is a cutting-edge generative model designed for video synthesis and enhancement through masked token modeling. Unlike traditional autoregressive video generation methods, MAGVIT predicts video frames in parallel using a spatial-temporal masked modeling approach, significantly improving efficiency and reducing computational complexity. It uses 3D vector quantization (VQ) to tokenize video data, allowing the model to learn temporal dependencies and contextual relationships across frames. MAGVIT has been applied to medical video generation, motion-based anomaly detection, surgical video synthesis, and pain assessment through facial expressions in healthcare (Nguyen et al. 2024a). This model's ability to generate high-quality and temporally consistent video sequences makes it particularly useful for predicting disease progression, reconstructing missing frames in medical imaging, and assisting in video-based diagnostic applications.

Yu et al. (2023) introduced MAGVIT as a multi-task generative video model, demonstrating its capability to handle various video generation tasks, including frame interpolation, inpainting, and anomaly detection. Their findings highlighted that MAGVIT achieves state-of-the-art video fidelity and efficiency performance, significantly outperforming diffusion models and Autoregressive Video Transformers in both speed and quality. Another research by Nguyen et al. (2024a) explored the application of Masked Autoencoder Transformers for video-based pain assessment, leveraging facial micro-expressions to determine pain levels in non-verbal patients. The study showed that MAGVIT, when integrated with deep learning-based facial analysis, improves pain recognition accuracy by capturing subtle facial cues often overlooked by conventional models. Additionally, MAGVIT's self-supervised learning framework allowed it to generalize across diverse medical video datasets, making it an adaptable solution for healthcare applications. These studies underscore MAGVIT's potential in medical video analysis, offering a robust framework for enhancing video-based diagnostics, monitoring patient conditions, and improving AI-assisted medical imaging workflows. However, scalability challenges, interpretability concerns, and computational demands remain key areas for further exploration in real-world clinical applications.

3.3.6 DALL-E

DALL-E and its subsequent iterations, DALL-E 2 and DALL-E 3, are advanced text-to-image generative models developed by OpenAI¹ that leverage transformer-based architectures to synthesize high-quality images from textual descriptions. These models employ diffusion-based and autoregressive approaches to progressively enhance image resolution, coherence, and realism. Their application in healthcare has been transformative, particularly in medical imaging, diagnostic assistance, and visual data augmentation. By generating synthetic medical images, these models support deep learning training datasets, simulate rare pathological conditions, and enhance AI-driven diagnosis. Additionally, DALL-E models facilitate the visualization of complex medical concepts, allowing physicians, educators,

¹ <https://openai.com>.

and researchers to create precise medical illustrations without relying on real patient data. Integrating latent diffusion techniques and contrastive image-text pretraining (CLIP) further refine image generation, improving contextual relevance and realism (Fezari et al. 2023; Borzunov et al. 2022).

Zhang and Shao (2024) explored OpenAI's DALL-E model, highlighting its potential to generate high-resolution medical images for diagnostic assistance. Their study demonstrated how DALL-E-generated synthetic datasets could significantly improve AI training models, particularly in radiology, dermatology, and pathology. While the generated images maintained visual fidelity, ensuring domain-specific accuracy and regulatory compliance remains challenging. Similarly, Fezari et al. (2023) explored the broader implications of generative AI, including DALL-E's contributions to healthcare. Their study examined the model's capacity to enhance medical education and clinical decision-making by generating realistic anatomical and pathological representations while also addressing ethical concerns related to bias in synthetic image generation and the potential misuse of AI-generated medical visuals. Additionally, Borzunov et al. (2022) investigated collaborative training strategies for large-scale transformer-based image generators like DALL-E. They developed an optimized distributed training infrastructure to reduce computational costs and improve accessibility for medical research institutions. Their findings suggested that DALL-E-based models could broaden access to high-quality medical imaging datasets, particularly in regions with limited medical imaging resources. These studies highlight the transformative role of DALL-E models in healthcare, from synthetic data generation for AI training to enhanced medical education and diagnostic visualization. However, challenges remain, particularly in ensuring data authenticity, mitigating ethical concerns, and improving domain-specific accuracy.

3.3.7 Perceiver and perceiver IO

Perceiver and perceiver IO are general-purpose architectures designed to efficiently handle high-dimensional and multimodal inputs. These models extend the capabilities of Transformers by introducing a latent space bottleneck, allowing them to process large-scale structured data with linear scalability. Traditional Transformers scale quadratically with input size, whereas Perceiver models use cross-attention and iterative processing to refine complex inputs into a lower-dimensional latent representation before further transformation. This structure enables them to handle diverse data types, including medical imaging, genomics, and multimodal patient records, without requiring domain-specific modifications (Jaegle et al. 2021a, b).

In healthcare, Perceiver models are beneficial for integrating heterogeneous clinical data, analyzing multimodal medical images, and supporting diagnostic decision-making. They enable more robust disease prediction and personalized treatment recommendations by efficiently fusing inputs from various modalities, such as EHR, medical imaging, and physiological signals. For instance, Jaegle et al. (2021b) introduced the Perceiver model, highlighting its ability to process high-dimensional structured inputs using iterative attention mechanisms. The authors demonstrated its superior performance in image classification and multimodal tasks, showcasing its potential adaptability to medical image analysis. They emphasized that the Perceiver's latent bottleneck significantly reduces computational complexity, making it a promising candidate for scaling deep learning models to large healthcare

datasets. Building on this foundation, Jaegle et al. (2021a) developed Perceiver IO, which extends Perceiver's capabilities by enabling structured and task-specific output generation. This architecture is particularly relevant in medical applications requiring complex output structures, such as segmentation of medical images, genomic sequence prediction, and clinical decision support systems. Their study demonstrated that Perceiver IO surpasses traditional transformer models in tasks such as language modeling and multimodal reasoning, making it well-suited for processing longitudinal patient records and clinical reports. In a domain-specific adaptation, Lee (2023) explored the application of Perceiver IO in high-fidelity facial texture reconstruction for medical imaging. Their research focused on generating 3D facial textures from in-the-wild images, a process valuable for reconstructive surgery planning and facial anomaly detection. The study revealed that Perceiver IO's ability to encode high-dimensional mesh vertices into a latent representation allows for detailed and efficient facial texture synthesis, surpassing conventional mesh-based models. These studies illustrate the transformative potential of Perceiver and Perceiver IO in healthcare, from enhancing medical imaging and multimodal integration to optimizing patient data analysis for predictive modeling. However, challenges remain, particularly in fine-tuning these models for domain-specific medical tasks and ensuring interpretability in clinical applications.

3.3.8 Summary and open issues

Transformers excel in capturing contextual relationships across modalities and have driven advances in synthetic clinical text generation, medical report drafting, and multimodal alignment of images and records. Shared strengths include scalability, flexibility across domains, and the ability to exploit large pretrained models. However, they face several challenges, such as (i) extreme data and compute requirements, (ii) risks of hallucination and factual inaccuracy, (iii) limited interpretability of attention maps, and (iv) privacy concerns when models memorize sensitive text or images. These recurring patterns are consolidated in Table 7. Future directions include smaller domain-specific Transformers, hallucination-mitigation techniques, and auditing frameworks to ensure trustworthy clinical deployment.

3.4 Diffusion models for synthetic data generation

Diffusion models generate data by iteratively denoising random noise through a probabilistic forward-reverse process. By progressively refining samples across hundreds or thousands of steps, they achieve state-of-the-art fidelity and diversity in image generation. In healthcare, diffusion approaches are rapidly gaining adoption for cross-modality medical image synthesis (e.g., MRI to CT), low-dose CT reconstruction, histopathology augmentation, and longitudinal signal forecasting. Variants include denoising diffusion probabilistic models (DDPMs), score-based models, and latent diffusion models, which reduce compute costs by operating in compressed latent spaces. Detailed applications and variants are described in the subsections (including Table 8) below.

Table 7 Transformer family—common strengths, use-cases, limitations, and future directions

Aspect	Strengths	Healthcare use-cases	Limitations and challenges	Future directions
Context modeling	Captures long-range dependencies via self-attention	Clinical text generation; EHR sequence modeling	Risk of hallucination; factual inaccuracy	Fact-checking modules; clinician-in-the-loop evaluation
Multimodality	Flexible integration across images, text, and signals	Radiology report generation; image-text retrieval	Alignment across modalities still immature	Multimodal pretraining with clinical supervision
Scalability	Leverages large pretraining datasets	Biomedical LLMs (e.g., PubMed-based) for text synthesis	High data and compute demand	Domain-specific lightweight Transformers; parameter-efficient tuning
Interpretability	Attention maps provide partial transparency	Identifying clinical features linked to text	Attention weights not always reliable	Causal interpretability tools; counterfactual attention analysis
Clinical fidelity	Generates fluent, structured narratives	Synthetic discharge summaries; patient-doctor dialogue	May omit or invent diagnoses; risk of bias	Evaluation benchmarks aligned with clinical correctness
Privacy	Trained on large-scale medical records	Surrogate datasets without sharing raw notes	Model memorization of PHI possible	Differentially private Transformers; data masking and audit pipelines

3.4.1 Conditional diffusion models

Conditional diffusion models (CDMs) are a powerful class of generative models in healthcare, offering enhanced flexibility for tasks such as medical image synthesis, time-series imputation, and data augmentation. CDMs incorporate conditioning information to guide denoising, enabling targeted generation based on input constraints. This property makes them particularly useful in healthcare applications requiring structured control over synthetic data generation, such as MRI reconstruction, super-resolution, and missing data imputation (Zhu et al. 2023; Chung et al. 2022).

Recent research highlights the efficacy of CDMs in diverse medical applications. Chung et al. (2022) introduced Come-Closer-Diffuse-Faster (CCDF), an acceleration framework for CDMs, optimizing the reverse diffusion process for faster convergence in medical image reconstruction. Their model demonstrated superior MRI reconstruction and image inpainting performance, reducing computational overhead while maintaining high reconstruction fidelity. Tashiro et al. (2021) developed Conditional Score-Based Diffusion Models for Probabilistic Time-Series Imputation (CSDI), significantly improving time-series imputation accuracy by capturing temporal dependencies in healthcare datasets. Their model reduced mean absolute error by up to 20%, demonstrating robust handling of missing medical data. Zhu et al. (2023) introduced Conditional Text Image Generation with Diffusion Models (CTIG-DM), applying CDMs to generate high-quality text images for medical documentation and handwriting recognition tasks. Their model improved the diversity and realism of generated text images, enhancing automated document processing in healthcare. Collectively, these studies underscore the growing role of CDMs in healthcare, offering advanced generative capabilities with improved control over output structures. While CDMs have demonstrated superior performance in handling complex medical data, challenges such as

Table 8 Comparison of diffusion models for synthetic data generation in healthcare

Variant	Key features	Supported data types	Limitations
Conditional Diffusion Models	Enables guided generation with structured control over output constraints; Useful for targeted medical data synthesis	Medical imaging; Time-series data; Handwritten medical documentation	Computationally intensive; Requires precise conditioning signals
Latent Diffusion Models	Operates in lower-dimensional latent space for efficient generation; Supports multimodal synthesis	High-resolution medical images; Neuroimaging; Medical text-to-audio data	Sensitive to latent space transformations; May require fine-tuning for different modalities
Progressive Distillation Diffusion Models	Reduces inference steps via knowledge distillation while maintaining generation fidelity; Optimized for real-time applications	Pathology image inpainting; Clinical text-to-speech; Drug discovery datasets	Trade-off between inference speed and output quality; Less effective for highly structured data
Adversarial Diffusion Models	Combines adversarial training with diffusion processes to improve robustness and sample diversity	Echocardiography; Blood cell image augmentation; Adversarially robust medical AI models	Computationally expensive; Balancing adversarial training with diffusion steps is complex
Transformer-Based Diffusion Models	Use transformer architectures instead of U-Nets for enhanced long-range dependencies and multimodal synthesis	Medical image segmentation; Motion prediction for rehabilitation; High-resolution medical imaging	High computational demand; Requires large-scale datasets for effective training
Manifold-Constrained Diffusion Models	Ensures generated data follows underlying manifold structures, improving clinical realism and reducing bias	Biomedical knowledge graphs; Disease progression modeling; Synthetic dataset balancing	Accurate manifold estimation is challenging; Requires domain-specific tuning
Autoregressive Diffusion Models	Models sequential dependencies via autoregressive conditioning; Effective for structured data generation	Multivariate medical time-series data; Patient monitoring datasets; Structured clinical records	Training is computationally expensive for long-sequence data; Suffers from slow generation speed
Discrete Diffusion Models	Adapts diffusion models for categorical and structured medical data, preserving clinical coherence	Electronic Health Records (EHR); Medical reports; Structured genomic sequences	Difficult to scale for multimodal data; Requires fine-tuned transition probabilities
Patch-Based Diffusion Models	Generates images in localized patches, improving efficiency and allowing fine-grained control over synthetic image quality	Radiology scans; MRI reconstruction; Small-scale medical datasets	Requires effective patch-stitching mechanisms to maintain global coherence
Energy-Based Diffusion Models	Integrates energy-based priors for structured and robust synthetic data generation; Improves model interpretability	High-resolution medical imaging; Adversarially robust biomedical models; Probabilistic healthcare AI	Training instability due to complex energy function estimation; Computationally intensive

computational efficiency, real-time applicability, and domain adaptation remain active areas of research.

3.4.2 Latent diffusion models

Latent diffusion models (LDMs) have emerged as a powerful generative technique for synthetic data generation in healthcare. They offer computational efficiency while maintaining high data fidelity. Unlike pixel-space diffusion models, LDMs operate in a lower-dimensional latent space, enabling more efficient training and inference. This makes them

particularly valuable for medical imaging tasks such as image synthesis, domain adaptation, and modality translation (Pinaya et al. 2022; Blattmann et al. 2023). LDMs also promise in multimodal applications, including medical text-to-image and text-to-audio generation (Liu et al. 2023).

Recent research has explored various applications of LDMs in healthcare, such as Blattmann et al. (2023) introduced a novel approach that extends LDMs to high-resolution video synthesis by aligning latent representations temporally, which could be applied to medical video data such as surgical recordings or dynamic radiology sequences. Their method demonstrated improved temporal consistency and image quality while maintaining computational efficiency. Liu et al. (2023) proposed AudioLDM, a latent diffusion framework for text-to-audio generation, which can be leveraged in healthcare applications such as diagnostic audio synthesis, patient monitoring, and speech disorder analysis. By encoding audio signals into a continuous latent space, AudioLDM allows for precise and computationally efficient generation, improving the quality of synthetic audio datasets for medical AI models. Pinaya et al. (2022) explored the use of LDMs in neuroimaging, particularly for synthesizing high-quality brain scans. Their study demonstrated how LDMs improve structural consistency in synthetic MRIs and aid in data augmentation for rare neurological conditions. By leveraging latent space transformations, their method enhanced the realism and variability of generated medical images, making LDMs a promising tool for neuroimaging research and clinical AI applications. These studies highlight the transformative potential of LDMs in synthetic data generation for healthcare. Their ability to generate high-quality, modality-specific data while reducing computational overhead positions them as a promising alternative to traditional generative models. However, challenges remain in optimizing latent representations for diverse medical domains and ensuring that the generated data aligns with real-world clinical distributions.

3.4.3 Progressive distillation diffusion model

Progressive distillation diffusion models (PDDMs) have emerged as an efficient alternative to traditional diffusion models by reducing the computational overhead associated with iterative generative processes. Unlike standard diffusion models that require numerous time steps to generate high-quality synthetic data, PDDMs leverage knowledge distillation techniques to accelerate sampling while maintaining generation fidelity. This property makes them particularly relevant in healthcare applications requiring real-time or near-real-time synthetic data generation, such as medical imaging, EHR synthesis, and molecular generation for drug discovery (Meng et al. 2023; Yin et al. 2024).

Recent studies have demonstrated the effectiveness of PDDMs in various medical applications. Meng et al. (2023) introduced a novel distillation framework for classifier-free guided diffusion models, significantly improving sampling efficiency. Their method reduced the required inference steps by 20x while maintaining visual quality comparable to traditional models. The study applied this approach to medical image inpainting and text-guided image generation, achieving high-fidelity reconstructions of pathology slides and radiological scans. Yin et al. (2024) proposed a Distribution Matching Distillation (DMD) approach that transforms a diffusion model into a one-step image generator. By minimizing an approximate KL divergence, their model successfully produced synthetic biomedical images with high structural consistency while reducing computational demands. Their

results demonstrated that the proposed method could accelerate synthetic data generation for histopathological image analysis without sacrificing diagnostic accuracy. Additionally, Huang et al. (2022) explored PDDMs in the context of speech synthesis for clinical applications, particularly in text-to-speech (TTS) models used for assistive technologies. Their model, ProDiff, directly parameterized clean data predictions instead of score estimation, significantly reducing refinement iterations. By employing progressive distillation, they achieved 24x faster speech synthesis with only two inference steps while maintaining naturalness in clinical speech datasets. This development is particularly beneficial in applications such as voice-based diagnostics and real-time medical communication tools. Overall, these studies highlight the potential of PDDMs in healthcare by offering an efficient, scalable, and high-fidelity approach to synthetic data generation. Their ability to optimize inference speed while preserving data integrity positions them as a valuable tool in medical AI applications. Recent advances have introduced acceleration strategies for diffusion, such as knowledge distillation and scheduler optimization, which significantly reduce the number of sampling steps. However, these gains come with trade-offs, such as distilled or accelerated models often show a modest drop in perceptual quality (e.g., lower FID or SSIM) compared to full-step counterparts. In medical imaging, early evaluations suggest that while acceleration maintains structural fidelity for common modalities (MRI, CT), subtle details relevant to diagnosis may be degraded, underscoring the need for domain-specific validation (Meng et al. 2023).

3.4.4 Adversarial diffusion models

Adversarial diffusion models (ADMs) integrate adversarial training mechanisms with denoising diffusion processes to improve synthetic data generation in healthcare. Unlike conventional diffusion models that rely solely on iterative denoising, ADMs introduce an adversarial component that enhances sample diversity, preserves high-quality features, and optimizes generation speed. This makes them particularly valuable for medical image translation, data augmentation, and adversarial robustness in healthcare AI models. By leveraging adversarial guidance, ADMs can generate high-fidelity medical images while maintaining structural consistency, making them useful for echocardiography synthesis, blood cell image augmentation, and adversarial training in clinical AI (Ngasa et al. 2024; Nguyen et al. 2024b; Chen et al. 2024a).

Recent studies have demonstrated the effectiveness of ADMs across various medical applications. Villaizán-Vallelado et al. (2024) propose a diffusion model tailored for tabular healthcare data imputation and generation, integrating a conditioning attention mechanism, transformer-based denoising, and dynamic masking. Their approach improves statistical fidelity while maintaining privacy by avoiding direct replication of real patient data. The model was evaluated against state-of-the-art techniques, such as Variational Autoencoders and Generative Adversarial Networks (GANs), demonstrating superior performance in machine learning efficiency, statistical similarity, and privacy risk mitigation. Meanwhile, Nguyen et al. (2024b) introduced Dataset Diffusion, a diffusion-based framework that synthesizes pixel-level semantic segmentation datasets for medical imaging applications. By leveraging self-attention exponentiation and class-prompt cross-attention, their method enhances segmentation quality while reducing the reliance on labor-intensive pixel-wise annotation. Benchmarked against standard datasets, their approach outperformed previous

methods, showcasing diffusion models' effectiveness in generating high-fidelity medical images with structured annotations. Recently, Ngasa et al. (2024) developed a diffusion-based WGAN-GP model for blood cell image augmentation, enhancing synthetic dataset quality and improving white blood cell classification accuracy to 95%. Additionally, Chen et al. (2024a) explored diffusion-based data augmentation to improve model performance in medical imaging tasks. Their study demonstrated that synthetic images generated by diffusion models could enhance classification accuracy in medical datasets, surpassing traditional GAN-based augmentation strategies. Beyond synthetic healthcare data generation, diffusion models are also studied for their detectability and impact on medical image integrity. Corvi et al. (2023) analyzed forensic traces left by adversarial diffusion models and assessed existing synthetic image detectors, highlighting their vulnerabilities in distinguishing diffusion-generated medical images from real ones. Their findings suggest that while diffusion models offer enhanced realism, they also introduce nuanced details that can be utilized for authenticity verification. Collectively, these studies emphasize the growing role of ADMs in healthcare, offering new pathways for privacy-preserving data synthesis, medical image augmentation, and forensic validation. While ADMs provide substantial improvements, challenges remain in optimizing computational efficiency and ensuring realistic domain adaptation in synthetic medical data.

3.4.5 Transformer-based diffusion models

Transformer-based diffusion models (TBDMs) extend diffusion processes by leveraging Transformers' self-attention mechanism, improving global feature learning and context-aware generation. Traditional diffusion models often rely on U-Net architectures, but TBDMs introduce Transformers as the primary backbone, enhancing long-range dependencies and hierarchical representations. This approach is particularly beneficial in healthcare applications such as medical image segmentation, motion prediction for rehabilitation, and scalable synthetic data generation (Tian et al. 2024). These models can handle multimodal medical data by incorporating Transformers, improving generalization across different imaging modalities while optimizing computational efficiency (Pan et al. 2023).

TBDMs have demonstrated significant potential for generating high-fidelity synthetic medical data, such as TransFusion, proposed by Sikder et al. (2023), is a diffusion-based generative model that integrates Transformers to synthesize long-sequence time-series data, addressing limitations in recurrent and convolutional architectures. TransFusion incorporates a transformer encoder within the backward diffusion process to enhance long-term temporal dependencies, significantly outperforming existing models like TimeGAN and GT-GAN in sequence prediction tasks. Through novel evaluation metrics that assess both the statistical and predictive quality of synthetic data, TransFusion ensures the generated time-series data maintains essential structural properties while improving model stability and scalability. Similarly, Pan et al. (2023) developed a Transformer-based Denoising Diffusion Probabilistic Model (TBDDPM) for 2D medical image synthesis, leveraging a Swin-transformer architecture to enhance image reconstruction across multiple imaging modalities, including chest X-rays, MRI, and CT scans. Their study introduced a novel training framework to improve the quality, authenticity, and diversity of synthetic images, with evaluations showing a substantial improvement in realism, as measured by FID and IS. Additionally, beyond standard synthetic image generation, diffusion-based transformers

have been evaluated for their ability to preserve critical medical biomarkers. Hosseini and Serag (2025) explored the use of diffusion models across diverse medical imaging domains, investigating whether generated synthetic images maintain key diagnostic features such as lung markings and retinal abnormalities. Their findings indicate that these models not only produce visually realistic images but also retain essential medical biomarkers, with classifiers trained on synthetic data achieving AUC scores between 0.8 and 0.99, highlighting their applicability in clinical decision support systems. Furthermore, their study emphasized the role of Swin-Vision Transformers in improving diffusion model performance by capturing global contextual features, making them particularly effective for high-dimensional medical imaging tasks. Collectively, these advancements illustrate the growing role of TBDMs in healthcare, providing innovative solutions for privacy-preserving synthetic data generation, disease diagnosis augmentation, and biomarker validation. However, challenges remain in optimizing computational costs and ensuring robust generalization across diverse medical datasets.

3.4.6 Manifold-constrained diffusion model

Manifold-constrained diffusion models (MCDMs) extend diffusion models by incorporating manifold-based constraints to improve the fidelity and structure of generated data. Standard diffusion models assume that data distributions follow unconstrained, high-dimensional spaces, which may lead to unrealistic synthetic samples. In contrast, MCDMs introduce constraints that align generated data with underlying lower-dimensional manifolds, ensuring that synthetic medical images, biomedical signals, and structured health data maintain clinically valid structures (De Bortoli 2022). This makes them highly relevant for medical image synthesis, bias mitigation in synthetic datasets, and efficient high-fidelity data augmentation in healthcare applications.

Recent work has examined how diffusion models relate to low-dimensional data geometry and how to guide sampling toward unbiased generations. De Bortoli (2022) provide theoretical convergence guarantees for denoising diffusion models when the target distribution lies on a low-dimensional manifold, giving quantitative W_1 bounds under the manifold hypothesis. Su et al. (2023) introduce Manifold-Guided Sampling (MCG), an unsupervised guidance that steers samples toward an estimated data manifold. Empirically, MCG yields consistent FID and sFID improvements across multiple datasets and schedulers and qualitatively produces a more balanced gender mix on CelebA-HQ, without reporting a numeric percentage of bias reduction. Together, these studies highlight the promise of manifold-aware diffusion, while open challenges remain in accurate manifold estimation, computational cost, and extending to multimodal healthcare data.

3.4.7 Autoregressive diffusion models

Autoregressive diffusion models (ARDMs) extend diffusion processes by incorporating sequential generative modeling, where each data component is generated one step at a time instead of in parallel. This approach allows ARDMs to model complex dependencies across medical data, making them particularly useful in multivariate time-series forecasting, medical image generation, and structured biomedical data modeling (Rasul et al. 2021). Unlike conventional diffusion models that simultaneously require iterative denoising across all

dimensions, ARDMs allow for an order-agnostic process where each element is modeled independently while still capturing conditional relationships (Kong et al. 2023).

Recent research has demonstrated the utility of ARDMs in various healthcare applications. Hoogetboom et al. (2021) introduced a general autoregressive diffusion framework that significantly reduces the number of diffusion steps required for synthetic data generation, improving efficiency while maintaining fidelity. Rasul et al. (2021) proposed Time-Grad, an autoregressive denoising diffusion model for multivariate probabilistic time series forecasting, which demonstrated state-of-the-art forecasting accuracy in real-world medical datasets, effectively modeling temporal dependencies in patient monitoring systems. Additionally, Kong et al. (2023) explored ARDMs for graph-based healthcare applications, introducing GRAPHARM, an autoregressive diffusion model that enables structured biomedical knowledge graph synthesis while ensuring connectivity and preserving clinical relationships. These studies highlight the robustness, adaptability, and efficiency of ARDMs in healthcare, but challenges remain in scaling ARDMs to high-dimensional multimodal datasets and optimizing computational performance.

3.4.8 Discrete diffusion models

Discrete diffusion models (DDMs) extend diffusion modeling to categorical and discrete data, enabling applications in medical coding, structured data synthesis, and EHR generation. Unlike traditional continuous diffusion models operating in high-dimensional real-valued spaces, DDMs employ discrete state-space transitions via categorical corruption and denoising processes. This ensures that generated data adheres to structured medical constraints, making DDMs particularly useful in EHR synthesis, structured biomedical data generation, and medical layout processing (Han et al. 2024). By leveraging a discrete-state transition mechanism, DDMs can capture categorical dependencies, ensuring that synthetic medical records maintain clinically meaningful distributions (Sun et al. 2022; Inoue et al. 2023).

Recent research highlights the effectiveness of DDMs in diverse healthcare applications such as Sun et al. (2022) introduced a score-based continuous-time discrete diffusion model, demonstrating that incorporating stochastic jump processes in Markov chains enhances categorical medical data generation. Inoue et al. (2023) developed LayoutDM, a discrete diffusion model for structured medical layout generation, proving its utility in controllable UI and medical form synthesis. Recently, Han et al. (2024) proposed EHR-D3PM, a diffusion model designed explicitly for synthetic EHR generation, achieving state-of-the-art fidelity and privacy preservation while enhancing disease classification accuracy in downstream tasks. These studies showcase DDMs' potential in privacy-preserving synthetic medical data generation. Still, challenges remain in scaling discrete models for multimodal healthcare applications and optimizing sampling efficiency in complex categorical datasets.

3.4.9 Patch-based diffusion model

Patch-based diffusion models (PBDMs) introduce a localized training paradigm that divides images into smaller patches, allowing faster training and better data efficiency in synthetic medical data generation. Unlike conventional diffusion models operating on full-scale images, PBDMs learn a conditional score function at the patch level, ensuring context-

aware synthesis across multiple scales (Wang et al. 2024b). This makes them particularly beneficial for medical image inpainting, data augmentation for small medical datasets, and anomaly detection in radiology scans (Wang et al. 2024a). These models reduce computational complexity by integrating patch-wise score matching while maintaining high-resolution synthetic images (Aversa et al. 2023).

Recent research demonstrates the effectiveness of PBDMs in various healthcare applications, such as DiffPatch, introduced by Wang et al. (2024b), is a diffusion-based customizable patch generation framework that enables the synthesis of adversarial patches for security applications in medical imaging and beyond. It employs Null-text inversion and Incomplete Diffusion Optimization (IDO) to maintain semantic coherence while optimizing for adversarial effectiveness. The model enhances patch realism while preserving attack efficiency, making it a powerful tool for controlled adversarial attacks in medical imaging. Meanwhile, DiffInfinite, proposed by Aversa et al. (2023), extends patch-based diffusion for histopathological image synthesis, enabling scalable, high-resolution medical image generation. Unlike conventional inpainting techniques, DiffInfinite leverages parallel random patch diffusion to ensure that synthesized histopathology images maintain long-range correlations and detailed cellular structures. Evaluation with pathologists and anti-copying metrics confirmed the method's potential in generating large histological datasets while preserving patient privacy. Additionally, Wang et al. (2024a) introduced Patch Diffusion, a training framework that reduces training time by 2x while maintaining comparable image generation quality, making it particularly beneficial for small-scale medical datasets. Collectively, these studies emphasize the impact of PBDMs in medical imaging, from histopathology augmentation to cross-modality synthesis, ensuring both privacy preservation and high-fidelity synthetic data generation. However, challenges remain in optimizing patch-wise training strategies and balancing local detail preservation with global coherence.

3.4.10 Energy-based diffusion models

Energy-based diffusion models (EBDMs) integrate energy-based modeling (EBM) with diffusion processes, enhancing structured controllability and robustness in synthetic data generation. Unlike conventional diffusion models that rely on explicit likelihood-based training, EBDMs learn an energy function that implicitly models the probability distribution of data, making them particularly useful for medical image generation, adversarially robust healthcare AI, and probabilistic biomedical data modeling (Yu et al. 2022b). These models improve sample diversity and interpretability by incorporating energy constraints, ensuring that generated medical data remains clinically coherent and structurally sound.

Recent research has demonstrated the effectiveness of EBDMs in various healthcare applications. Du et al. (2023) proposed a compositional generation framework that utilizes EBDMs and MCMC-based sampling to enhance medical image synthesis while reducing computational complexity. Similarly, Yu et al. (2022b) introduced a diffusion-amortized MCMC approach, which improves latent space exploration and convergence stability, addressing challenges in multimodal medical data generation. Recently, Guo et al. (2023) developed EGC, a unified diffusion energy-based model for simultaneous image classification and generation, demonstrating superior classification accuracy and adversarial robustness on biomedical datasets. These studies highlight EBDMs' potential in robust,

high-fidelity synthetic healthcare data generation, though challenges persist in scalability, computational efficiency, and training stability.

3.4.11 Summary and open issues

Diffusion models have established themselves as leading generative frameworks for medical imaging, consistently outperforming GANs on perceptual metrics such as FID, SSIM, and PSNR. Their training is generally more stable, with reduced risk of mode collapse, and they support fine-grained image fidelity that is critical for diagnostic realism. However, diffusion models face important challenges, such as (i) slow sampling times due to iterative refinement, (ii) high computational requirements, (iii) limited interpretability of denoising steps, and (iv) a lack of standardized clinical evaluation benchmarks beyond perceptual metrics. These shared patterns are summarized in Table 9. Ongoing research is focused on accelerated sampling strategies, hybrid diffusion-transformer architectures, and clinically informed evaluation pipelines to ensure utility in diagnostic and regulatory settings.

3.5 Federated learning for generative AI models

Federated learning enables collaborative model training across multiple institutions without sharing raw data, thereby addressing rigorous privacy and regulatory constraints in healthcare. When combined with generative models such as GANs or VAEs, FL allows the generation of synthetic datasets that reflect diverse, multi-center populations while maintaining data locality. This approach has been applied to multi-site MRI/CT harmonization, synthetic EHR generation, and histopathology augmentation across hospitals. Recent innovations include federated GANs (FedMed-GAN), federated VAEs, and distributed synthetic learning (DSL), which improve cross-site fidelity and robustness. Detailed implementations and case studies are described in the subsections (including Table 10) below.

Table 9 Diffusion model family—common strengths, use-cases, limitations, and future directions

Aspect	Strengths	Healthcare use-cases	Limitations and challenges	Future directions
Fidelity and diversity	Superior perceptual realism; reduced mode collapse vs GANs	MRI to CT synthesis; dermatology; histopathology	Requires large training sets	Semi-supervised or weakly supervised diffusion
Training stability	Robust convergence; less sensitive to adversarial dynamics	Radiology augmentation; CT denoising	Computationally intensive training	Lightweight or distilled diffusion models
Low-dose / reconstruction	Iterative refinement improves denoising and resolution	Low-dose CT, PET super-resolution	Long sampling time hinders deployment	Accelerated sampling (DDIM, score distillation)
Cross-modality generation	Works well for unpaired modality translation	MRI to CT, multi-modal harmonization	Risk of anatomical distortion	Structural priors; anatomy-aware losses
Clinical fidelity	High-fidelity detail in tissues and lesions	Rare lesion generation; longitudinal progression	Lack of clinical ground-truth benchmarks	Clinician-in-the-loop evaluation standards
Privacy	Less prone to overfitting than GANs	Surrogate datasets in multi-institutional imaging	Still vulnerable to inversion attacks	Differentially private diffusion; federated diffusion training

Table 10 Comparison of federated generative models in healthcare applications

Variant	Key features	Supported data types	Limitations
Federated Conditional GAN	Enables privacy-preserving GAN training in decentralized settings; Enhances model generalization across healthcare institutions	Medical imaging; EHRs; Disease risk prediction	High communication costs; Sensitive to non-IID data distribution
Federated Variational Autoencoder	Integrates VAE with federated learning for privacy-preserving feature extraction and synthetic data generation	Medical imaging; IoT patient monitoring; Genomic data	Computationally intensive; Slower convergence in non-IID settings
Federated Generative Models for Edge Computing	Deploys generative models across edge devices, reducing dependency on centralized computing	Real-time patient monitoring; Medical wearables; ICU telemetry	Energy consumption constraints; Potential trade-offs in model performance
Federated Multimodal GANs	Facilitate multimodal learning across decentralized institutions, enabling cross-modality medical image synthesis	MRI-CT translation; Histopathology-genomics integration	Requires advanced privacy techniques (e.g., differential privacy) for regulatory compliance
Federated 3D GAN	Optimized for 3D medical image generation and segmentation; Preserves privacy in multi-institutional collaborations	CT; MRI; 3D radiology; Anatomical modeling	High storage and computational requirements; Increased model complexity
Federated Disentangled Representation Learning	Separates global and local variations in decentralized medical data for improved interpretability	MRI anomaly detection; Treatment effect estimation; Genomic analysis	Computationally expensive; May require large-scale federated fine-tuning
Federated Cross-Silo Generative Models	Focused on collaboration across high-resource institutions such as hospitals and research labs, ensuring secure multi-party learning	Cross-institutional medical imaging; multi-source predictive modeling	High communication overhead; Complex model synchronization

3.5.1 Federated conditional generative adversarial networks

Federated conditional generative adversarial networks (FedCGAN) is a privacy-preserving approach that enables synthetic data generation while maintaining decentralized learning. Traditional GANs require centralized data, whereas FedCGAN allows multiple healthcare institutions to train a model collaboratively without exposing patient data (Majeed and Hwang 2024). This approach is particularly valuable in scenarios where data privacy regulations, such as HIPAA and GDPR, restrict the centralization of medical records. FedCGAN utilizes a generator-discriminator framework where each participating client locally trains a model using its dataset, and only model updates (not raw data) are aggregated into a global model. This ensures secure data sharing while benefiting from diverse data distributions across institutions.

FedCGAN has been applied to tasks such as medical image synthesis, disease risk prediction, and anomaly detection, improving model robustness across different healthcare facilities. For instance, a significant development is the Conditional Tabular GAN based federated model, proposed by Majeed and Hwang (2024), which enhances tabular synthetic data generation using a class-conditioned generator. This approach effectively models mixed-type data distributions, including both numerical and categorical attributes, while ensuring that synthetic data maintains statistical properties similar to real-world datasets. The study highlights the importance of balancing privacy and utility, showing that FedC-

GAN achieves competitive performance in machine learning tasks while reducing privacy risks associated with direct data sharing. Experimental comparisons with state-of-the-art generative models reveal that FedCGAN surpasses traditional federated GANs by preventing mode collapse, improving diversity in synthetic data, and ensuring data fairness across distributed nodes. Additionally, Maliakel et al. (2024) introduced FLIGAN, a variant of FedCGAN designed to address incomplete data scenarios in federated settings. By generating synthetic samples for underrepresented classes, FLIGAN significantly improved model accuracy by 20% over traditional FL baselines. Collectively, these studies reinforce the role of Federated Conditional GANs in enabling privacy-compliant synthetic data generation for healthcare and other regulated domains, facilitating collaborative AI advancements without compromising data confidentiality. However, challenges such as non-IID (non-independent and identically distributed) data distributions, computational overhead, and communication efficiency remain areas of active research.

3.5.2 Federated variational autoencoders

Federated variational autoencoders (FedVAE) integrate variational autoencoders (VAE) with FL to enable privacy-preserving and distributed machine learning in healthcare. Traditional VAEs are widely used for latent space modeling, feature extraction, and synthetic data generation. Still, their integration with FL allows multiple institutions to collaboratively train models without sharing raw patient data, ensuring compliance with privacy regulations such as HIPAA and GDPR. In this approach, each participating node, such as hospitals, medical centers, or IoT devices, locally trains a VAE-based model, and only latent representations or model updates are transmitted to a global aggregator (Bhatti and Choi 2024). This process enhances data privacy and security and enables better generalization across diverse healthcare datasets. FedVAE has shown promise in remote patient monitoring, anomaly detection, medical image synthesis, and EHR analysis. Particularly in IoT-based healthcare, it allows hospitals to train predictive models collaboratively without exposing sensitive patient data, enhancing diagnostic accuracy while maintaining privacy. The method is especially effective in heterogeneous, non-IID environments, where data distributions vary significantly across healthcare providers. To address data imbalance and slow convergence, FedVAE incorporates adaptive aggregation strategies, clustering techniques, and latent space transformations, ensuring efficient learning across diverse healthcare settings.

Findings from the literature reinforce FedVAE's effectiveness, such as Pfizner and Arnrich (2022) introduced Federated Variational Autoencoder with Differentially-Private Decoder (DPD-fVAE), which enhances privacy by synchronizing only the decoder component in FL. This approach reduces privacy leakage risks while maintaining data utility, as the encoder remains private and user-specific. By incorporating differential privacy, DPD-fVAE mitigates privacy concerns without severely degrading the quality of the generated synthetic data. Experimental results on MNIST, Fashion-MNIST, and CelebA datasets demonstrate that DPD-fVAE achieves competitive performance in classification tasks and synthetic data resemblance, as measured by FID, while reducing the privacy cost per epoch compared to traditional federated VAEs. Another approach, Synthetic Data Sharing (SDS), proposed by Apellániz et al. (2025), leverages FL to improve synthetic data generation in heterogeneous healthcare environments. SDS contrasts with conventional federated averaging (FedAvg) by enabling the sharing of locally generated synthetic patients across nodes,

thereby enhancing data diversity and reducing disparities between data-rich and data-poor institutions. Their study integrates Variational Autoencoder with Bayesian Gaussian Mixture Models (VAE-BGM), demonstrating that SDS significantly improves synthetic data quality in non-IID scenarios. Their experiments on healthcare datasets show that SDS outperforms FedAvg by reducing divergence in synthetic data distributions by over 50% in the worst-performing nodes. This research underscores SDS as an effective FL strategy to address data scarcity and heterogeneity in healthcare while preserving patient privacy. Additionally, Bhatti and Choi (2024) proposed an IoT-driven FedVAE framework to enhance data aggregation in smart healthcare applications, leveraging autoencoders to capture latent feature representations. Their approach resulted in 20.8% higher accuracy and 7% lower loss compared to conventional methods. These studies highlight FedVAE's role in privacy-preserving, distributed learning, offering improved model performance, data security, and communication efficiency. However, challenges remain, particularly in computational complexity, federated model drift, and optimizing aggregation strategies.

3.5.3 Federated generative models for edge computing

Federated generative models for edge computing (FedGM-EC) integrate FL with generative AI to enable distributed, privacy-preserving data generation across edge devices. Unlike centralized generative models, which pose privacy risks and communication overhead, FedGM-EC ensures that data remains decentralized, mitigating risks in healthcare settings where patient confidentiality is paramount. By training generative models across distributed edge nodes, FedGM-EC eliminates the need for data transfer to a central server. Instead, it relies on model updates or latent representations for aggregation, ensuring data security and compliance with privacy regulations. These models facilitate medical data augmentation, missing data imputation, anomaly detection, and patient-specific model customization, making them particularly beneficial for resource-constrained healthcare environments (Li et al. 2024). Additionally, their ability to support heterogeneous and non-IID data improves model generalizability, ensuring robustness in diverse clinical applications (Xiong et al. 2023). Leveraging adaptive aggregation strategies, knowledge distillation, and hierarchical FL techniques, FedGM-EC enhances training efficiency, mitigates data imbalance issues, and optimizes communication bandwidth.

Xiong et al. (2023) introduced a hierarchical federated generative model for multi-source heterogeneous data processing in IoT-based medical systems, demonstrating a 30% improvement in model accuracy for synthetic MRI and CT scans while ensuring patient privacy. Li et al. (2024) proposed the FIMI (Filling the Missing) framework, leveraging generative AI for data augmentation in FL across heterogeneous edge devices, reducing energy consumption by up to 50% while enhancing global model accuracy in non-IID environments. Meanwhile, Little et al. (2023) conducted a scoping review on FL for synthetic data generation, identifying GAN-based approaches as the predominant method, highlighting the ongoing challenge of quantifying privacy risks. These studies collectively underscore the role of FedGM-EC in privacy-preserving, AI-driven healthcare solutions. However, challenges such as high computational demands, communication constraints, and the need for robust personalization techniques remain barriers to adoption.

3.5.4 Federated multimodal GANs

Federated multimodal generative adversarial networks (FedMM-GANs) leverage FL to facilitate secure and decentralized training of generative models on multimodal medical datasets. FedMM-GANs are particularly beneficial for multimodal medical data, including MRI, CT scans, genomic sequences, and histopathological images, where data-sharing restrictions often hinder centralized deep learning applications. By distributing learning across institutions, FedMM-GANs ensure that models benefit from diverse datasets without compromising patient confidentiality. These models are instrumental in cross-modality medical image synthesis, missing data imputation, and enhanced disease prognosis by integrating different imaging and clinical data types (Chang et al. 2023). Recent advancements integrate privacy-preserving techniques such as differential privacy and homomorphic encryption to further protect sensitive patient information while enhancing generative model performance across heterogeneous data distributions (Chen and Pan 2024).

Several studies have explored the potential of FedMM-GANs in healthcare applications, such as Wang et al. (2023b) introduced FedMed-GAN, a federated domain translation model designed for cross-modality brain image synthesis. This model effectively addresses data heterogeneity across institutions by balancing paired and unpaired neuroimaging data during training. Their study demonstrated that FedMed-GAN outperformed state-of-the-art centralized models while mitigating mode collapse, a common challenge in generative modeling. Chang et al. (2023) introduce Distributed Synthetic Learning (DSL), a novel framework leveraging a central generator and multiple distributed discriminators to enable synthetic data generation while addressing data heterogeneity and missing modalities. DSL enhances FL by integrating multimodality learning, missing modality completion, and continual learning capabilities. Evaluations on multi-center datasets, including cardiac CTA, brain tumor MRI, and histopathology nuclei images, demonstrate that DSL outperforms traditional FL models such as FedMed-GAN and AsynDGAN in segmentation and classification tasks. The introduction of the Dist-FID metric further enables robust synthetic image quality assessment, surpassing existing FID-based evaluations. Another key advancement in FedMM-GAN is the development of AsynDGAN, an asynchronous distributed GAN framework that ensures data privacy while learning from multimodal and heterogeneous datasets. AsynDGAN employs a central generator that captures joint data distributions from multiple institutions while allowing each site to maintain local discriminators, thus reducing bandwidth overhead and privacy risks. Unlike conventional federated GANs, AsynDGAN supports missing modality completion, enabling robust synthesis of absent imaging modalities. Experimental results on the BraTS dataset validate its effectiveness in generating synthetic MRI modalities and improving segmentation accuracy (Chang et al. 2020). By mitigating challenges such as communication inefficiency and data heterogeneity, these approaches illustrate the potential of FedMM-GAN in multi-institutional collaborations, enhancing model generalization while safeguarding patient privacy.

3.5.5 Federated 3D GAN

Federated 3D generative adversarial networks (Fed3DGAN) have emerged as a significant innovation in the healthcare domain, particularly in medical imaging applications. These models integrate FL with 3D GAN architectures to enable collaborative training across

multiple healthcare institutions while preserving patient privacy. The primary architectural components of Fed3DGAN include a global generator model that learns representations from multiple client nodes, a discriminator trained on local datasets, and a federated averaging mechanism that aggregates model updates from different sites. These models excel in synthesizing high-quality 3D medical images, such as CT and MRI scans, aiding in the augmentation of limited datasets, improved segmentation, and enhanced diagnostic accuracy (Li et al. 2020a; Suzuki 2024).

The reviewed studies demonstrate the efficacy of Fed3DGAN in various medical imaging tasks. Li et al. (2020a) proposed a federated simulation framework for 3D cardiac CT synthesis that integrates a physics-driven generative approach. This method utilizes two learnable modules: one for synthesizing 3D cardiac shapes and materials and another for CT volume rendering. The disentangled geometry-material modeling enables cross-institutional training while preserving privacy, significantly improving segmentation performance across multiple datasets. Another study by Peketi et al. (2023) introduces FLWGAN (Federated Learning with Wasserstein GANs) for brain tumor segmentation, leveraging a modified 3D WGAN-GP to address mode collapse and improve convergence stability. By employing a 3D-UNet with an attention module for enhanced segmentation accuracy, this framework demonstrates state-of-the-art performance on the Medical Segmentation Decathlon (MSD) dataset across multiple tumor segmentation tasks. The results indicate that FL with GAN-based synthetic data augmentation mitigates data scarcity issues while maintaining data security. Furthermore, Suzuki (2024) developed Fed3DGS (Federated 3D Gaussian Splatting), a scalable federated framework for large-scale 3D medical reconstruction. By incorporating appearance modeling and a distillation-based model update scheme, Fed3DGS efficiently reconstructs 3D medical images while adapting to non-IID data variations, making it suitable for long-term multi-institutional collaborations. These studies highlight the potential of Fed3DGANs in medical imaging by enabling robust and privacy-preserving collaborative learning. Integrating federated architectures with 3D generative models enhances the synthesis of high-fidelity medical images, thereby supporting diagnostic improvements, model generalization, and equitable access to high-quality AI-driven healthcare solutions across multiple institutions.

3.5.6 Federated disentangled representation learning

Federated disentangled representation learning (FedDis) enhances FL by ensuring privacy-preserving training while addressing heterogeneous data distributions in medical applications. Unlike conventional FL models, which struggle with variability across institutions, FedDis learns disentangled latent representations, allowing it to separate global and local feature variations. This enables better generalization across medical datasets while maintaining personalized adaptability for each participating node. Using variational inference and structured latent space learning, FedDis ensures that distinct factors of variation are preserved, leading to improved interpretability and robustness in healthcare AI models (Almansoori et al. 2024).

Liu et al. (2022a) propose a disentangled representation learning framework for medical imaging that isolates content (task-relevant factors) from style (domain-specific variations). Their approach enhances robustness across different imaging modalities and mitigates biases introduced by domain shifts in federated settings. The framework incorporates VAEs and

Normalizing Flows to enforce disentanglement, ensuring that latent factors retain clinical relevance while remaining invariant to acquisition-specific distortions. Experiments on MRI and CT datasets demonstrate that this method significantly improves diagnostic accuracy in cross-domain medical applications compared to standard FL approaches. Complementing this, Almansoori et al. (2024) introduce PaDPaF (Partial Disentanglement with Partially-Federated GANs), a federated generative model that separates global content representations from client-specific style variations. Using a contrastive regularization term based on Barlow Twins, their model ensures that federated content features generalize well across clients while preserving local personalization through private style encoders. PaDPaF's federated GAN framework demonstrates improved performance in privacy-preserving data synthesis, domain adaptation, and synthetic data augmentation for healthcare applications. By enabling cross-institutional collaborations without data sharing, these approaches highlight the potential of FedDis in enhancing data security, model personalization, and robustness in distributed healthcare AI. However, challenges remain in optimizing scalability, computational efficiency, and adaptive feature learning.

3.5.7 Federated cross-silo generative models

Federated cross-silo learning (FCSL) has emerged as a promising solution for multi-institutional medical AI applications. Unlike traditional FL paradigms, which often involve many low-power devices, cross-silo FL focuses on a small number of organizations, such as hospitals and research institutions, that possess significant computational resources and require secure collaboration while maintaining privacy (Zhao and Huang 2023; Ali et al. 2022). This paradigm is fundamental in healthcare, where data sharing across institutions is restricted due to ethical, legal, and regulatory constraints. FCSL allows these institutions to train deep learning models collaboratively while keeping patient data local, mitigating privacy risks. However, challenges such as non-IID data distribution, communication efficiency, and model heterogeneity pose significant hurdles.

Recent studies have introduced innovative approaches for federated cross-silo generative models to address these issues. SiloFuse, introduced by Shankar et al. (2024), leverages latent tabular diffusion models to generate synthetic data while preserving privacy in vertically partitioned datasets. By employing autoencoders to encode sensitive features into latent representations, SiloFuse ensures that original data remains undisclosed while capturing cross-silo feature correlations through distributed diffusion-based synthesis. Experimental evaluations demonstrate that SiloFuse outperforms GAN-based approaches in both data resemblance and utility scores, offering a scalable and communication-efficient alternative for federated synthetic data generation. Similarly, Tran et al. (2024) propose a Personalized Privacy-Preserving Federated Learning (PPPFL) framework, which integrates Differentially Private Generative Adversarial Networks (DP-GANs) with Model-Agnostic Meta-Learning (MAML). This approach not only generates privacy-preserving synthetic data but also enhances personalization in FL by addressing non-IID data distribution challenges. PPPFL empirically outperforms standard FL baselines, demonstrating improvements in classification accuracy across multiple datasets. Another notable development is DPSDA-FL, proposed by Abacha et al. (2024), which employs foundation models for differentially private synthetic data generation in FL environments. Unlike traditional GAN-based solutions, DPSDA-FL utilizes pre-trained diffusion models to generate high-quality synthetic

data, mitigating the risks of training instability and mode collapse often associated with GANs. Experimental results on the CIFAR-10 dataset reveal that DPSDA-FL significantly improves global model recall and classification accuracy, effectively addressing data heterogeneity challenges in federated settings. Additionally, another study explored auxiliary data augmentation, where public or synthetically generated datasets are used to mitigate heterogeneity among clients, further optimizing performance (Ali et al. 2022). Collectively, these approaches highlight the growing role of federated generative models in facilitating secure and collaborative data sharing, particularly in regulated domains such as healthcare. By integrating differential privacy mechanisms, latent encoding techniques, and foundation models, federated cross-silo generative frameworks are enhancing the accessibility, security, and utility of synthetic data in distributed machine learning applications. Despite these advancements, challenges remain, particularly in optimizing aggregation mechanisms, reducing communication overhead, and ensuring fairness across clients.

3.5.8 Summary and open issues

Federated generative models combine the privacy advantages of FL with the flexibility of generative synthesis, making them highly attractive for clinical collaborations. Their shared strengths include compliance with privacy regulations (HIPAA, GDPR), the ability to pool rare disease data across centers without centralization, and support for synthetic surrogate datasets for benchmarking. However, challenges remain, such as (i) non-IID data distributions across sites reduce model performance, (ii) communication overhead complicates scaling, (iii) potential leakage through gradients persists, and (iv) standardized evaluation protocols for privacy and fidelity are lacking. These common themes are consolidated in Table 11. Future work should explore federated diffusion models, secure aggregation pro-

Table 11 Federated generative models—common strengths, use-cases, limitations, and future directions

Aspect	Strengths	Healthcare use-cases	Limitations and challenges	Future directions
Privacy and compliance	Data remain local; aligns with HIPAA/GDPR	Multi-institutional MRI/CT harmonization; EHR synthesis	Gradient leakage risk	Secure aggregation; DP in federated updates
Rare disease pooling	Enables collaboration without sharing raw cases	Rare tumor imaging across hospitals	Non-IID data adversely affects convergence	Domain adaptation; personalization strategies
Cross-site generalization	Exposure to diverse data distributions improves robustness	Histopathology augmentation; cross-scanner harmonization	Performance drops on heterogeneous sites	Adaptive weighting; site-specific fine-tuning
Communication efficiency	Scales collaboration beyond pairwise sites	Multi-hospital imaging federations	High communication and compute overhead	Model compression; efficient communication protocols
Evaluation standards	Supports synthetic surrogate datasets for audits	Benchmarking of clinical models with synthetic data	Lack of standardized metrics for privacy and utility	Unified evaluation combining fidelity, utility, and privacy
Integration with generative models	Extends GANs/VAEs to federated settings	Federated GANs (FedMed-GAN); DSL; federated VAEs	Fidelity lower than centralized training	Federated diffusion; hybrid FL architectures

protocols, and audit frameworks that balance fidelity, privacy, and efficiency for real-world multi-institutional deployments.

4 Discussion

Generative AI has emerged as a transformative technology in healthcare, enabling synthetic data generation for medical imaging, EHRs, biomedical signal processing, and drug discovery. Section 3 systematically analyzed four major categories of generative models, including GANs, VAEs, Transformers, and Diffusion models, alongside their respective variants. Each model demonstrates unique capabilities, trade-offs, and challenges when applied to different healthcare domains. This section critically compares these models across performance metrics, strengths and weaknesses, suitability for diverse medical data types, and training complexity, followed by practical considerations for real-world deployment and key challenges that must be addressed for broader clinical adoption.

4.1 Comparative analysis across models

This section provides a comprehensive comparative analysis of major generative AI models discussed in this review, including GANs, VAEs, Transformer-based models, Diffusion models, and Federated generative approaches. It critically evaluates these models across four key dimensions: performance metrics, strengths and weaknesses, suitability for various healthcare data types, and training complexity. This comparative assessment highlights the unique advantages, limitations, and trade-offs associated with each model class, offering practical insights into their deployment potential across diverse healthcare applications. To provide a visual comparative overview based on the synthesis of our review findings, we present a qualitative assessment of major generative model families across critical performance dimensions (Fig. 2). The summary of best-suited model types for common use-case priorities, such as realism, speed, diversity, fine-tuning flexibility, out-of-the-box usability, and privacy-preserving training, is presented in Table 12 to complement Fig. 2.

4.1.1 Performance metrics

Generative models vary significantly in their performance metrics, influencing their effectiveness in healthcare applications. GANs, particularly StyleGAN, and ProGAN, are known for producing high-fidelity synthetic medical images with fine-grained details, making them highly suitable for radiology and histopathology (Fetty et al. 2020; Golfe et al. 2023). However, models such as CycleGAN and AttnGAN emphasize domain adaptation and image translation, improving generalization across different imaging modalities such as MRI and CT scans (Lakshmi et al. 2023; Breen et al. 2021; Ma et al. 2018). VAE-based models, including standard VAEs and ConVAEs, are widely used in medical imaging and EHR synthesis, offering stable training and well-structured latent representations. However, standard VAEs may produce slightly blurry images due to their probabilistic nature (Singh and Ogunfunmi 2021; Berahmand et al. 2024). Transformer-based generative models, such as DALL-E and ViT, exhibit superior contextual awareness and multimodal data integration, excelling in generating synthetic text and enhancing medical image classification (Han et al. 2022;

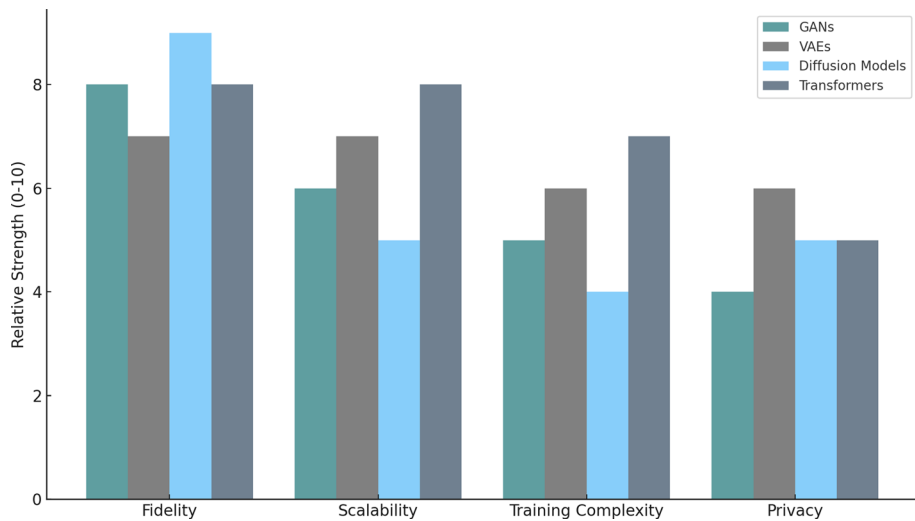


Fig. 2 Comparative strengths of generative models in healthcare

Table 12 Best-suited generative models based on use-case priorities

Use-case priority	Best-suited model type	Example models	Notes
High realism (fidelity)	Diffusion Models (Pinaya et al. 2022; Zhu et al. 2023; Breen et al. 2021)	LDMs, ADMs, PDDMs	Best for high-resolution medical imaging; highest fidelity but computationally expensive
Fast training and inference	VAEs (Fallahian et al. 2024; Biswal et al. 2021; Nikolentzos et al. 2023; Zhu et al. 2021, 2020; Han et al. 2020)	Stacked VAE, CVAEs	Low resource requirement; suitable for EHR and structured data synthesis
Variability / Diversity	GANs (Breen et al. 2021; Karras et al. 2019; Sauer et al. 2023)	StyleGAN2, CycleGAN	Excellent diversity in image generation; potential risk of mode collapse
Out-of-the-box usability	Transformer Models (Li et al. 2024; Shah-Mohammadi and Finkelstein 2024; Zhang and Shao 2024)	DALL-E, GPT variants	Pre-trained models available for text and multimodal tasks; minimal domain adaptation required
Easy fine-tuning	Autoencoders (Fallahian et al. 2024), Transformers (Han et al. 2022)	Sparse VAE, ViT	Good for low-data regimes and customization; lower overfitting risk
Privacy-preserving training	Federated Models (Pennisi et al. 2022; Yang et al. 2023)	FedCGAN, FedVAE, FedMM-GAN	Best for distributed, multi-institutional training; challenged by non-IID data

Zhang and Shao 2024). Federated generative models, including FedCGAN and FedVAE, maintain privacy preservation and decentralized model convergence, ensuring secure training across multiple institutions while facing challenges in handling heterogeneous, non-IID medical data distributions (Pennisi et al. 2022; Yang et al. 2023). Diffusion models, such as LDMs and ADMs, are emerging as leading candidates for generating high-fidelity synthetic data while ensuring controlled structural integrity. However, they require extensive computational resources and longer inference times compared to other models (Pinaya et al.

2022; Wang et al. 2023c). PDDMs attempt to mitigate this by accelerating inference speeds, making them more viable for real-time applications (Meng et al. 2023). Each generative model type demonstrates strengths and trade-offs regarding fidelity, diversity, realism, and stability, influencing their suitability for different healthcare tasks.

To complement the qualitative synthesis in Fig. 2 and the use-case mapping in Table 12, we summarize representative quantitative results across domains in Section 4.1.5 (Table 13). For instance, StyleGAN and Diffusion Models consistently achieve higher fidelity in medical imaging tasks (as reflected by lower FID and higher precision scores in respective studies), whereas VAEs offer superior reconstruction and latent control in EHR generation. Figure 2 summarizes these trade-offs visually, highlighting how model capabilities vary across fidelity, scalability, and privacy dimensions.

4.1.2 Strengths and weaknesses

Despite their advantages, each generative model category exhibits inherent strengths and weaknesses, affecting their deployment in healthcare applications. GANs are well-known for their high-resolution image generation and are extensively used for data augmentation and rare disease detection. However, many variants, including WGANs and ACGANs, suffer from training instability, mode collapse, and sensitivity to hyperparameters, limiting their applicability in complex, high-dimensional medical data scenarios (Welfert et al. 2024; Tomar and Gupta 2023; Liu et al. 2019; Saxena and Cao 2021; Saad et al. 2024). VAEs, particularly Stacked VAEs, and Sparse VAEs, excel in feature extraction, noise removal, and structured data synthesis, making them useful for EHR augmentation and medical anomaly detection (Fallahian et al. 2024). However, their generative quality often lacks the diversity seen in GAN-based models. Transformer-based generative models, such as GPT-variants, excel in clinical text generation and medical literature summarization, improving EHR documentation efficiency. Still, these models often suffer from hallucination and domain-adaptation challenges due to the lack of medically focused datasets (Xie et al. 2023; Rashidi et al. 2024; Liu et al. 2022c; Xiao et al. 2024). Federated generative models are essential for ensuring privacy-preserving AI in healthcare by enabling secure multi-institutional collaboration. Yet, their training overhead and communication costs remain a significant challenge, especially for FedGM-EC and FedMM-GANs (Liu et al. 2024b; Nguyen et al. 2022; Li et al. 2025; Qayyum et al. 2022). Diffusion models, while providing state-of-the-art realism and controlled generative processes, require significant computational resources and often longer inference times, limiting real-time deployment in medical imaging and EHR generation (Liu et al. 2024a; Yang et al. 2024). The evolution of TBDMs attempts to mitigate these weaknesses by integrating self-attention mechanisms, improving efficiency and multimodal adaptability (Tian et al. 2024). Each model demonstrates specific strengths tailored to different healthcare domains, but none provides a universal solution for all generative tasks.

4.1.3 Suitability to healthcare data types

The suitability of generative models for different healthcare data types varies significantly depending on their architectural design and data-handling capabilities. Medical imaging tasks, including MRI reconstruction, histopathology augmentation, and cross-domain image translation, are best handled by GANs (StyleGAN, BigGAN, and CycleGAN) and Diffu-

Table 13 Representative quantitative benchmarks across domains and model families

Domain	Task and dataset(s)	Model family: Specific model	Metrics and reported values	Key findings
Imaging (Pan et al. 2024)	MRI to CT synthetic CT generation (Brain and Prostate MRI/CT pairs)	Diffusion: MC-DDPM (3D Transformer-guided diffusion)	Brain: MAE 43.3 HU, PSNR 27.0, SSIM 0.965; Prostate: MAE 60 HU, PSNR 26.9, SSIM 0.849	Diffusion outperforms GAN baselines on fidelity; higher compute cost
Imaging (Wang et al. 2023b)	Cross-modality MRI synthesis (IXI, BraTS2021)	GAN: FedMedGAN (CycleGAN + DP-SGD, federated)	PSNR 28.3 (Fed) vs 27.1 (Centralized); SSIM 0.989 vs 0.986	Federated training improves fidelity while preserving DP guarantees
Histopathology (Xue et al. 2021)	Cervical epithelium and lymph node classification (patch-level)	GAN: HistoGAN (conditional GAN with selective filtering)	AUROC +6.7 pp (cervical), +2.8 pp (lymph node) vs baseline	Selective GAN augmentation boosts classifier performance; naive augmentation less effective
Imaging (Yang et al. 2025)	CT and MRI cross-modality synthesis (seg2med benchmark)	Diffusion: Score-based	SSIM 0.94 (CT), 0.89 (MR); FID 3.6 (CT), 6.5 (MR)	Diffusion achieves high realism, outperforming GANs on FID and SSIM
Ophthalmology (Li et al. 2020b)	Fundus to fluorescein angiography (FFA) image translation; HRA (hospital) and Isfahan MISP (paired)	cGAN (PatchGAN discriminator) and saliency loss	Paired image-to-image translation; on Isfahan MISP, PSNR 25.16 dB, SSIM 0.8268 (vs CycleGAN 19.65 dB, 0.5799; pix2pix 23.43 dB, 0.7438)	Saliency loss boosts small-vessel and leakage fidelity; outperforms CycleGAN/pix2pix on both PSNR and SSIM
Dermatology (Carrasco Limeros et al. 2022)	Dermoscopy lesion synthesis and augmentation; ISIC 2019/2020	StyleGAN2-ADA (unconditional per class and conditional), including federated setup	Central and federated training; best centralized conditional run reported FID $\approx$ 7.02, KID 0.24%, P 0.75, R 0.44; augmentation changed classifier AUC only by a few tenths of a pp (e.g., baseline AUC 98.8%, aug $\sim$ 98.6–98.8%)	High visual fidelity (low FID) but only marginal downstream gains on large ISIC sets; federated setup converges faster and helps the smallest site
Structured EHR (Yoon et al. 2023)	MIMIC-III and eICU EHR synthesis	Hybrid: EHR-Safe (encoder–decoder and GAN)	Downstream accuracy gap <3% vs real; KS drift <0.03; MIA success $\approx$ chance	Synthetic EHR preserves utility and offers near-ideal privacy
Structured EHR (Han et al. 2024)	Phenotyping tasks (MIMIC-III)	Diffusion: Discrete D3PM	AUROC boost 0.02–0.05 (guided augmentation); Lower membership inference risk	Discrete diffusion improves task performance while lowering privacy risk
Time-series ECG (Adib et al. 2025)	Beat-level ECG synthesis	GAN: Comparison of 5 GANs (WGAN, TimeGAN, etc.)	TSTR AUROC 0.89 (synthetic) vs 0.91 (real); DTW distance low	GANs can preserve morphology; TimeGAN best balance of fidelity and utility
EEG (Ling et al. 2022)	Sleep stage classification (single-channel EEG)	GAN: RDB-DCGAN	Minority-class F1 boost significantly; overall staging F1 $\sim$ 0.87 vs 0.89 (real)	Synthetic EEG augmentation improves class balance and staging performance

Table 13 (continued)

Domain	Task and dataset(s)	Model family: Specific model	Metrics and reported values	Key findings
Clinical Text (Ren et al. 2025)	MIMIC-IV clinical letters	Transformer: Synthetic4Health (masked LM generation)	BERTScore >0.90 (light masking), <0.90 (noun masking); NER F1 comparable to real	Template-based strategies enable realistic clinical notes with low PHI risk
Radiology report generation (Tanida et al. 2023; You et al. 2021; Chen et al. 2022)	Radiology image–report pairs (IU-Xray)	Region-guided Transformer (vision encoder and text decoder)	CLIP score 0.78, BLEU-4 0.34; FID ~22	Region guidance improves localization faithfulness and text metrics over prior encoder-decoder baselines
Multi-center imaging–federated-style synthesis (Chang et al. 2023)	Mining multi-center heterogeneous medical data with DSL across radiology/pathology sites	DSL (privacy-preserving GAN-like pipeline)	Abstract reports up to ~55% segmentation improvement under limited labels; DSL outperforms FLGAN/ AsynDGAN/ FedMed-GAN; shows lower memorization risk and practical deployment at 20 hospitals	DSL better handles cross-site heterogeneity than conventional FL methods, improving fidelity and downstream tasks without sharing raw data
Privacy-preserving EHR (Jordon et al. 2018)	Tabular health registry synthesis	GAN: PATE-GAN	$\epsilon = 2$ DP; AUROC drop <0.03 vs non-DP	PATE-GAN achieves formal DP with minimal utility loss

sion models (LDMs and CDMs) due to their ability to generate high-resolution, structured medical images (Pinaya et al. 2022; Zhu et al. 2023; Breen et al. 2021; Karras et al. 2019; Sauer et al. 2023; Brock et al. 2019; Chung et al. 2022). Autoencoder-based models, such as VGAEs and CVAEs, are instrumental in structured EHR augmentation and biomarker discovery, where well-defined latent representations improve data compression and feature extraction (Biswal et al. 2021; Nikolentzos et al. 2023; Duan et al. 2024; Friedman et al. 2025). Transformers, including GPTs and DALL-E variants, are essential for NLP-driven tasks, such as clinical text summarization, medical question-answering, and patient documentation automation (Li et al. 2024; Shah-Mohammadi and Finkelstein 2024; Zhang and Shao 2024). FL models, including FedMM-GANs and Fed3DGANs, are particularly suitable for distributed medical imaging tasks that require multi-institutional collaboration while maintaining data privacy compliance (Wang et al. 2023b; Murmu et al. 2024; Li et al. 2020a; Suzuki 2024). Time-series physiological signal modeling, such as ECG and EEG synthesis, benefits from RVAEs and ARDMs, which capture temporal dependencies and sequential structures effectively (Yang et al. 2024; Casolaro et al. 2023). Similarly, discrete generative models, such as DDMs, best suit structured EHR generation and categorical medical data synthesis, ensuring coherence in structured clinical records (Sarkar et al. 2024). The specific healthcare domain heavily influences the choice of model, required data fidelity, and computational feasibility.

Based on the comparative analysis presented, GANs (particularly StyleGAN and CycleGAN) are best suited for medical imaging tasks due to their ability to generate high-resolution and structurally coherent synthetic images. VAEs and their variants, such as ConVAE

and Sparse VAE, are most appropriate for structured EHR and tabular data, offering stable latent representations and good reconstruction. Transformer-based models are ideal for text-based healthcare tasks, including clinical summarization and documentation. Diffusion models stand out in generating high-fidelity multimodal data, particularly in imaging, though their computational cost is a limiting factor. For federated environments, FedGANs and FedVAEs are promising, but require further advances in handling non-IID data and communication efficiency. These findings can guide application-specific model selection in real-world healthcare settings.

4.1.4 Training complexity

The training complexity and resource requirements of the generative models determine their feasibility in real-world clinical environments. GANs, despite their remarkable performance in medical imaging, suffer from unstable training, mode collapse, and high computational costs, particularly for high-resolution models like BigGAN and StyleGAN (Bhat and Nanjundegowda 2025; Karras et al. 2019). While Autoencoders are more computationally efficient, models such as Stacked VAEs and Sparse VAEs require deep hierarchical structures, increasing their memory footprint and convergence time (Zhu et al. 2020, 2021; Han et al. 2020). Transformers, particularly GPT and ViT models, demand large-scale pre-training datasets and high memory consumption, making them impractical for low-resource healthcare settings without significant infrastructure investment (Hemmati et al. 2025; Tu et al. 2024). Federated generative models, including FedCGAN and FedVAE, introduce communication overhead and non-IID training challenges, limiting their scalability in real-time clinical deployments (Liu et al. 2024c; Wen et al. 2022). Diffusion models, though capable of producing high-fidelity synthetic data, require prolonged training durations and expensive inference procedures, restricting their adoption in resource-limited environments (Carlini et al. 2023; Shang et al. 2023; Meng et al. 2023). PDDMs attempt to accelerate inference by reducing computational steps, making them more viable for real-time applications (Meng et al. 2023; Yin et al. 2024). Similarly, PBDMs optimize generative efficiency by processing localized image regions, reducing overall training demands while maintaining quality (Wang et al. 2024a). However, for institutions with limited GPU capabilities, simpler architectures such as VAEs may provide a more feasible alternative for structured medical data generation. Each generative model category presents unique computational demands and resource constraints, influencing its adoption and scalability in healthcare AI applications.

4.1.5 Quantitative comparison across model families

To ground the comparative discussion in data, we compiled representative benchmarks from peer-reviewed studies across major domains (imaging, structured EHR, clinical text, and physiological time-series). Table 13 consolidates quality/utility metrics (e.g., PSNR/SSIM, FID/IS, AUROC/accuracy/MAE) and available privacy indicators (e.g., membership-inference AUC, k-anonymity, distance-to-closest-record). Overall, diffusion models currently lead image-to-image synthesis on PSNR/SSIM, while GAN- and hybrid approaches dominate selective augmentation effects in histopathology. For structured EHR, recent frame-

works report < 3% downstream utility gaps to real data while approaching ‘chance-level’ privacy attack success.

Across medical imaging, diffusion-based sCT synthesis outperforms GAN baselines on MAE/PSNR/SSIM (e.g., MAE 43 HU brain; SSIM 0.965) but at higher compute cost. In histopathology, selective GAN augmentation yields consistent accuracy gains ($\approx +2.8$ to 6.7 pp). For EHR synthesis, EHR-Safe reports <3% utility gap to real data for mortality prediction and near-ideal privacy attack rates. Clinical-text synthesis is emerging, early results show ROUGE-/BLEU-level similarity with manual PHI checks. Time-series augmentation (EEG/ECG) improves minority-class performance but remains sensitive to overfitting and evaluation choice, hence TSTR-style tests and attack-aware audits are recommended.

4.2 Practical considerations for healthcare deployment

The successful deployment of generative AI models in healthcare depends on multiple practical considerations, including structured real-world deployment strategies, data availability, ethical implications, regulatory compliance, model interpretability, computational feasibility, and integration into clinical workflows.

4.2.1 Data availability and quality

One of the most critical factors is data availability and quality, as generative models rely heavily on large-scale, high-quality datasets for effective training. GAN-based models require extensive labeled medical images to avoid mode collapse and poor generalization (Li et al. 2021; Welfert et al. 2024). Similarly, Transformer-based models need access to massive textual datasets for clinical document generation, often requiring domain-specific pretraining on medical literature or EHRs (Yenduri et al. 2024; Luo et al. 2022; Cho et al. 2024). FL-based generative models, such as FedMM-GAN and Fed3DGAN, attempt to address data access limitations by allowing institutions to collaborate without sharing raw patient data. Yet, challenges remain in handling non-IID data distributions and heterogeneous institutional datasets (Huang et al. 2024; Liu et al. 2024c). Diffusion models demand diverse and high-resolution datasets to generate clinically valid synthetic images. Still, their reliance on long training cycles and extensive data augmentation makes real-world implementation challenging (Wang et al. 2023c; Cao et al. 2024). Given these data constraints, multi-institutional collaborations and synthetic data augmentation strategies are essential for ensuring that generative models perform reliably across different patient demographics and clinical settings.

4.2.2 Model interpretability

Model interpretability and explainability play a crucial role in determining whether generative models can be trusted for clinical decision support and medical research. Autoencoder-based models, such as β -VAE and CVAE, are widely used for disease classification and structured data synthesis due to their transparent feature extraction process, making them easier to interpret than black-box generative models like GANs and diffusion models (Diallo et al. 2021; Pawar and Attar 2020; Eltager et al. 2023). VGAEs provide structured representations of disease progression and drug interactions, offering valuable insights for preci-

sion medicine (Lin et al. 2023; Liu et al. 2024b). Conversely, Diffusion models, including PDDMs and TBDMs, struggle with interpretability due to their complex multi-step generative processes, making them harder to validate in critical healthcare applications (Daneshfar et al. 2024; AlSaad et al. 2024). Self-attention mechanisms in Transformer-based models provide some level of interpretability by highlighting relevant textual or visual features. Still, challenges remain in applying these models for regulated medical AI applications (Tian et al. 2024). Developing explainable AI frameworks tailored for generative models ensures clinicians can trust and validate AI-generated outputs before clinical adoption.

4.2.3 Computational feasibility

Computational feasibility and resource constraints also affect the adoption of generative AI models in real-world healthcare settings. GANs and Diffusion Models, such as BigGAN, CycleGAN, and DDMs, require high-end GPUs and extensive parallelization for training, making them resource-intensive and costly for many hospitals and medical institutions (Bougueffa et al. 2024). Transformer-based generative models demand significant pretraining efforts, often requiring cloud-based computing infrastructure to achieve scalable AI deployments (Yenduri et al. 2024; Latibari et al. 2024). FL models introduce additional communication and synchronization overhead, impacting real-time performance in clinical settings (Mora et al. 2024). Institutions deploying AI-driven healthcare solutions must assess cost-effective strategies, such as cloud-based AI services and lightweight edge computing solutions, to optimize generative model performance within existing clinical workflows (Zhang et al. 2024).

4.2.4 Structured real-world deployment

Successful real-world deployment of generative AI models in healthcare necessitates a structured validation framework, involves several key steps. First, external validation must be performed using independent, heterogeneous datasets to verify model generalization across diverse populations and institutions (Yu et al. 2022a; Ramspek et al. 2021). Second, continual monitoring mechanisms should be established post-deployment to detect model drift, emergent biases, or degradation in performance over time (Nandan Prasad 2024). Third, randomized clinical trials (RCTs) are essential to evaluate the clinical efficacy, safety, and ethical acceptability of AI interventions in real-world environments (El Arab et al. 2025). In the context of FL environments, these strategies require specific adaptations. External validation must occur in a decentralized manner without compromising data privacy, continual monitoring should leverage federated auditing protocols, and federated RCTs may be necessary, wherein institutions collaboratively assess interventions without centralizing sensitive patient data. Developing structured, federated-compatible deployment pathways is crucial to build trust, ensure accountability, and achieve sustainable integration of generative AI solutions into healthcare systems.

4.2.5 Clinical workflows

Seamless integration into clinical workflows is a key consideration for the practical deployment of generative AI in healthcare. While GAN-based models and Diffusion models excel

in synthetic data augmentation for AI training, their integration into real-time diagnostic tools, medical imaging pipelines, and EHR systems remains challenging (Ibrahim et al. 2024; Mamo et al. 2024). VAE-based models, particularly RVAEs, have demonstrated strong applicability in longitudinal patient monitoring and predictive modeling, making them easier to integrate into existing hospital information systems (Duan et al. 2024). FL-based models, such as Fed3DGAN and FedMM-GANs generative models, are particularly well-suited for multi-institutional research collaborations, but their real-time deployment in clinical settings requires highly optimized model aggregation protocols (Arbaoui et al. 2024). Transformer-based models, including BART-based models, have already been integrated into clinical NLP applications, enabling efficient EHR summarization, automated reporting, and medical literature synthesis (Islam et al. 2023; Goyal et al. 2023; Guan et al. 2021).

4.3 Challenges and limitations

Despite the transformative potential of generative AI in healthcare, several challenges and limitations restrict its widespread adoption. One of the most pressing concerns is data fidelity and clinical reliability (Giuffrè and Shung 2023). While generative models have demonstrated significant advancements in synthetic medical data generation, ensuring that the generated data maintains clinical accuracy and preserves essential medical patterns remains a major challenge (Nahid and Hasan 2024). GANs, for instance, suffer from mode collapse, where the generator produces limited data variations, reducing diversity in synthetic samples (Zhang et al. 2018). This is particularly problematic in medical imaging and disease classification, where precise variability is crucial for training robust AI models (Arnold et al. 2011). Similarly, VAEs struggle with blurry reconstructions, which can impact the usability of synthetic medical images for diagnostic training (Singh and Ogunfunmi 2021; Berahmand et al. 2024; Molnár and Tamás 2024). Transformer-based models, such as GPTs, often suffer from hallucination, where they generate clinically inaccurate or misleading medical reports, posing risks in automated medical documentation (Xie et al. 2023; Xiao et al. 2024). Additionally, diffusion models face challenges related to inference speed and computational overhead, making real-time medical AI applications difficult (Ma et al. 2024). FL-based generative models, such as FedCGAN and FedVAE, introduce further complexities regarding model convergence and data consistency across decentralized training nodes (Pennisi et al. 2022; Yang et al. 2023). Addressing these fidelity issues requires improved evaluation frameworks, robust quality control mechanisms, and human-in-the-loop verification for generative AI outputs (Parker and Karbstein 2023; Akbar and Conlan 2024; Zhang et al. 2022).

Another significant limitation is computational cost and scalability (Wang et al. 2023a). Generative models, particularly high-resolution GANs, Transformer-based architectures, and Diffusion models, require massive amounts of computational power and storage resources, making them less accessible for resource-constrained healthcare environments. Training generative models on large-scale multimodal medical datasets (e.g., MRI-CT translation, histopathology-genomics integration, multi-source EHR synthesis) demands substantial GPU/TPU clusters, leading to high operational costs (Bhuyan et al. 2025). FL-based approaches introduce additional computational overhead due to communication synchronization between distributed nodes, limiting real-world feasibility in low-resource

clinical settings (Brisimi et al. 2018). Furthermore, training complex generative models on highly heterogeneous healthcare datasets, such as genomic sequences, multi-institutional medical records, and high-resolution 3D radiology scans, requires extensive hyperparameter tuning and optimization strategies to prevent overfitting, mode collapse, and computational inefficiencies (Ferdous and Efat 2023; Chen et al. 2024b). Additionally, ensuring model transparency and providing clinician-interpretable explanations remain critical for adoption, as emphasized by recent surveys in medical imaging (Mir and Rizvi 2025).

Data scarcity and bias (Bae et al. 2025; Hastings 2024) remain persistent challenges in generative AI applications for healthcare. While generative models are designed to mitigate data scarcity by producing synthetic datasets, issues related to bias, underrepresentation, and domain generalization persist (Draghi et al. 2024). Potential biases and weaknesses can arise at multiple levels. Data biases include sampling bias, demographic underrepresentation, and distributional shifts across institutions. Algorithmic biases involve model overfitting to dominant patterns, failure to generalize to minority subpopulations, and sensitivity to adversarial perturbations. User interaction biases may occur during model validation or clinical deployment, where human feedback loops or selective usage patterns could reinforce or mask certain outcomes. Addressing these diverse biases is essential to ensure fair, robust, and ethically responsible deployment of generative AI systems in healthcare settings. GANs, VAEs, and Diffusion models trained on skewed datasets may unintentionally reinforce existing demographic and diagnostic biases in AI-driven healthcare applications. For instance, StyleGAN-generated dermatology datasets may not accurately represent diverse skin tones, leading to biased AI-driven skin lesion diagnosis models (Thong et al. 2023). Similarly, Transformer-based generative models trained on clinical notes from a limited demographic may not generalize well to broader patient populations, leading to disparities in AI-assisted medical decision-making (Denecke et al. 2024). FL-based generative models, while offering privacy advantages, often struggle with heterogeneous data distributions across hospitals, impacting model generalization (Caldarola et al. 2022). Addressing bias and data scarcity challenges requires careful curation of training datasets, fairness-aware model training methodologies, and robust domain adaptation techniques to improve generalization across diverse patient populations. Additionally, future research should also focus on developing robust defenses against model inversion attacks and membership inference threats that could potentially exploit shared generative models in decentralized healthcare environments.

Compliance with privacy and ethical regulations remains a central challenge. While frameworks such as HIPAA (U.S.) and GDPR (EU) provide foundational safeguards, recent regulatory initiatives go further in explicitly addressing AI systems that may incorporate synthetic data. The EU Artificial Intelligence Act (2024) categorizes most medical AI systems as high-risk, requiring rigorous measures for data governance, documentation of synthetic datasets, risk management, and post-market monitoring (Lognoul 2025). In the United States, the FDA's Good Machine Learning Practice (GMLP) draft guidance (2021) emphasizes dataset representativeness, transparency of provenance, and independent validation, noting that synthetic data may be acceptable in device submissions when rigorously justified (Food et al. 2021). Recent FDA Digital Health initiatives (2023) further highlight the importance of bias auditing and clinical relevance testing when synthetic data augment real-world clinical trial evidence (Food et al. 2025). Similarly, the UK MHRA and NICE recognize synthetic data as useful in exploratory analyses and simulation studies, while cau-

tioning that it cannot yet replace real patient data in critical regulatory submissions (Medicines and products Regulatory Agency (MHRA) 2022). At a global level, the OECD AI Principles (2019) and the WHO Guidance on Ethics and Governance of AI in Health (2021) stress fairness, accountability, and human oversight in the use of synthetic data (OECD 2019; Thiebes et al. 2021; Guidance 2021). Collectively, these positions indicate that while regulators increasingly acknowledge the value of synthetic data, they expect it to serve as a supplementary tool, which is useful for development, validation, and augmentation, but not yet sufficient as a sole evidence-based for clinical approval.

Beyond governance and compliance, practical deployment must address concrete attack surfaces for generative models in healthcare. First, gradient leakage in federated or collaborative settings can enable reconstruction of training data from shared updates. The evaluations by Zhao et al. (2020) show that even partial gradients can expose sensitive information. Defenses such as secure aggregation (Truex et al. 2019) and homomorphic encryption (Fang and Qian 2021) mitigate this but increase communication and computational cost. Second, output leakage occurs when models memorize or reproduce training samples. Differential privacy (DP-SGD and its extensions) remains the leading defense, though tighter privacy budgets can reduce model fidelity (Bu et al. 2020; Tramer and Boneh 2020). Third, auxiliary-data attacks, including membership and attribute inference, leverage side information to test data presence or infer sensitive attributes. Frameworks such as Private Aggregation of Teacher Ensembles (PATE) have been adapted to healthcare, showing lower leakage at the expense of extra model training (Jayaraman and Evans 2019). More recently, evidence of training-data extraction from large generative models has motivated pre-release privacy audits and hybrid defenses that combine DP with cryptographic aggregation (Hu et al. 2022). In practice, these defenses introduce explicit privacy-utility-cost trade-offs that should be reported alongside quality and task metrics to support safe clinical adoption.

4.4 Future directions

Rapid advancements in generative AI for healthcare have highlighted numerous opportunities for future research and innovation. To fully harness the potential of GANs, VAEs, Transformers, Diffusion models, and FL-based generative models, several critical areas must be explored to enhance model performance, scalability, interpretability, ethical compliance, and clinical integration. Future research must address the current challenges of generative AI while improving its adaptability to real-world healthcare applications.

4.4.1 Fidelity and realism of synthetic data

One of the most promising directions is improving the fidelity and realism of synthetic medical data. While GANs, Diffusion models, and VAEs have demonstrated remarkable capabilities in generating high-resolution medical images, synthetic EHRs, and multimodal healthcare data, ensuring clinical validity and diagnostic reliability remains a key priority (Brock et al. 2019; Pan et al. 2023; Yoon et al. 2023; Pezoulas et al. 2024). Future research should focus on hybrid models that integrate multiple generative paradigms, such as combining Transformer-based architectures with diffusion models to enhance long-range dependencies in synthetic medical sequences or leveraging adversarial training with manifold

constraints to refine the realism of synthetic images in pathology and radiology. Additionally, self-supervised learning techniques can improve data representation quality, allowing models to generate clinically coherent outputs with minimal supervision (Shurrab and Duwairi 2022). Recent advancements in energy-based models, such as EBDMs and hierarchical latent space learning, offer promising pathways to refine generative outputs, ensuring that synthetic medical data aligns with real-world clinical distributions (Du et al. 2023; Cui et al. 2023).

4.4.2 Optimization of generative AI models

Another significant future direction is optimizing generative AI models for computational efficiency and scalability (Jabri et al. 2022). Training large-scale models, such as BigGAN, StyleGAN, ViTs, and Latent Diffusion models, requires substantial GPU/TPU resources, making them inaccessible to smaller healthcare institutions with limited computational infrastructure (Abdurahman et al. 2024). Future research should explore efficient model compression techniques (Tao et al. 2022), including quantization, pruning, and knowledge distillation, to reduce computational costs without compromising performance. Moreover, hierarchical diffusion models and autoregressive generative frameworks should be optimized for real-time applications, enabling faster medical image reconstruction, EHR augmentation, and real-time patient monitoring. Efforts to integrate FL with generative AI should also focus on decentralized model training optimizations, reducing communication overhead in multi-institutional healthcare collaborations (Gabrielli et al. 2023). Edge AI solutions for generative models should also be explored, particularly in remote and low-resource settings, enabling on-device synthetic data generation for real-time clinical decision support (Yousuf and Mahmud 2024).

4.4.3 Models' robustness and fairness

Another substantial research avenue is enhancing model robustness, fairness, and bias mitigation (Barbierato et al. 2022). Current generative models struggle with demographic and diagnostic biases, leading to ethical concerns in AI-assisted healthcare (Sharma et al. 2020). Future work should focus on fairness-aware training methodologies that ensure diverse representation across patient populations. One potential solution is domain adaptation and adversarial debiasing techniques, where models are trained across diverse clinical datasets to mitigate systemic biases (Breen et al. 2021; Correa et al. 2024). FedDis has shown promise in addressing domain shifts in distributed healthcare AI (Bercea et al. 2021), and future work should refine these models for improved generalization across multi-institutional datasets. Additionally, self-supervised contrastive learning could enhance representation learning in multimodal medical datasets, ensuring that generative models effectively capture disease variations across different patient demographics (Min et al. 2022). Future research should explore adaptive model calibration techniques to ensure fairness in AI-generated disease prediction, anomaly detection, and clinical report synthesis (Li and Jha 2024). It should also integrate explainable AI strategies to ensure clinically meaningful and trustworthy interpretations of synthetic data, consistent with the recommendations of Mir and Rizvi (2025).

4.4.4 Multimodal generative modeling

One of the key frontiers for generative AI in healthcare is multimodal generative modeling and cross-disciplinary applications (Ko et al. 2022; Ziegler et al. 2022). Generative AI models should evolve to handle complex, multi-source medical data, including medical imaging, genomic sequences, wearable sensor data, and unstructured clinical notes (Sheng et al. 2024). Transformer-based architectures, such as Perceiver IO, should be refined to handle heterogeneous healthcare modalities, improving the integration of multimodal patient records for comprehensive disease modeling (Jaegle et al. 2021a). LDMs and DDMs should also be optimized for multimodal fusion tasks, allowing AI systems to synthesize high-quality medical reports from combined text, image, and time-series data (Pinaya et al. 2022; Blattmann et al. 2023; Sun et al. 2022). Future work should also explore cross-disciplinary collaborations between AI researchers, healthcare professionals, and biomedical scientists to advance generative AI applications in drug discovery, protein structure modeling, and precision medicine. For instance, VGAEs (Nikolentzos et al. 2023) could be leveraged to model complex disease-gene interactions, accelerating biomarker discovery and personalized treatment development. Additionally, transfer learning remains underexplored in the context of generative healthcare AI. Cross-modal transfer approaches, such as CT2US for ultrasound kidney segmentation (Song et al. 2022), illustrate how synthetic data and transfer learning can be combined to address domain shifts and limited labeled data. Future work should more systematically evaluate these strategies for medical imaging, EHR, and multimodal synthesis.

4.4.5 Clinical applicability

While generative models have demonstrated state-of-the-art performance in synthetic data generation, their clinical applicability remains underexplored. Real-world clinical validation and translational AI research must be prioritized (Ibrahim et al. 2024). Future research should emphasize collaborations with hospitals, pharmaceutical companies, and regulatory bodies to validate generative AI models in real-world healthcare settings (Zamzmi et al. 2024; Ibrahim et al. 2024). Clinical trials assessing the effectiveness of synthetic medical data in augmenting AI-driven diagnostic systems, medical education, and telemedicine applications should be conducted to establish clinical trust in generative AI (Azizi et al. 2021). Furthermore, human-AI collaboration frameworks should be developed, ensuring that clinicians and healthcare practitioners can actively participate in AI-driven workflows, validate generative outputs, and fine-tune model predictions for personalized patient care (Wang et al. 2020; Reddy 2024).

5 Conclusion

Generative AI has revolutionized synthetic data generation in healthcare by addressing challenges related to data privacy, accessibility, and heterogeneity, enabling high-fidelity medical datasets for applications in medical imaging, EHR augmentation, disease modeling, and predictive analytics. This study provided a comprehensive analysis of four major categories of generative AI models, such as GANs, VAEs, Transformers, and Diffusion Models, includ-

ing Federated Learning for several generative models, highlighting their methodologies, applications, comparative advantages, and limitations. While these models significantly overcome healthcare AI challenges, including high-resolution image synthesis, structured data representation, and privacy-preserving learning, they still face barriers such as training instability, computational complexity, data bias, and regulatory constraints that hinder widespread clinical adoption. Future research must prioritize model robustness, scalability, and fairness, integrating hybrid generative approaches, domain adaptation, and explainable AI frameworks to bridge the gap between research advancements and real-world applications. Additionally, the development of privacy-preserving AI, including Federated Learning and secure synthetic data generation, will be critical for fostering multi-institutional collaborations while maintaining patient confidentiality. As generative AI continues to evolve, interdisciplinary collaboration among AI researchers, healthcare professionals, and policy-makers will be essential to ensure its responsible deployment, ultimately driving advancements in patient care, medical research, and healthcare innovation.

Acknowledgements This work is co-funded by the HORIZON.2.1 - Health Programme of the European Commission, Grant Agreement number: 101095661 - Innovative applications of assessment and assurance of data and synthetic data for regulatory decision support (INSAFEDARE).

Author Contributions Hafiz Muhammad Waseem is responsible for Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Visualization, and Writing – Original Draft Preparation. Saif Ul Islam is responsible for Conceptualization, Investigation, Methodology, Validation, Visualization, and Writing – Original Draft Preparation. Nikolaos Matragkas is responsible for Formal Analysis, Funding Acquisition, Project Administration, Validation, and Writing – Review & Editing. Gregory Epiphaniou is responsible for Conceptualization, Funding Acquisition, Investigation, Project Administration, Resources, Supervision, Validation, and Writing – Review & Editing. Carsten Maple is responsible for Formal Analysis, Funding Acquisition, Investigation, Project Administration, Resources, Supervision, Validation, and Writing – Review & Editing. Theodoros N. Arvanitis is responsible for Funding Acquisition, Project Administration, Validation, Visualization, and Writing – Review & Editing.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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